

REASONING & COMPUTER APTITUDE

- 1. Directions:** In the question given below, a passage is given followed by five statements. Read the information and answer the question that follows.

Recent research definitively proves that consuming at least three cups of black coffee daily prevents type 2 diabetes. Therefore, health experts recommend that everyone should drink three cups of black coffee per day to prevent diabetes.

Statements:

- I. The research indicating the benefits of coffee was funded by major coffee companies.
 - II. High consumption of coffee can lead to increased anxiety and sleep disorders.
 - III. Not everyone can tolerate coffee due to its caffeine content.
 - IV. Studies show that diet and exercise are more effective in preventing type 2 diabetes.
 - V. Excessive caffeine intake can have adverse health effects.
- Which of the following will weaken the above statement?

- (a) Only I
- (b) Only II
- (c) Only III
- (d) Only IV
- (e) Only V

- 2. Directions:** In the question given below, a passage is given followed by five statements. Read the information carefully and answer the question that follows.

Zenith Pharmaceuticals, a global leader in vaccine development, has been the primary supplier of vaccines to Country A's national immunization program for the past five years. Their contract is due for renewal in the next quarter. Meanwhile, NovaGen Biotech, an emerging competitor with innovative vaccine technology, is eager to secure the contract. The Ministry of Health in Country A is deliberating whether to renew the contract with Zenith or consider NovaGen as a new supplier.

Statements:

- I. Recent clinical trials have revealed that NovaGen's vaccines are more effective against emerging viral strains compared to those produced by Zenith.
- II. Zenith Pharmaceuticals has been accused of unethical practices, including data manipulation in clinical trials, by international health organizations.
- III. NovaGen Biotech's production facilities have failed multiple regulatory inspections due to non-compliance with safety standards.
- IV. The cost of vaccines offered by Zenith is significantly lower than those offered by NovaGen, easing the financial burden on Country A's healthcare budget.
- V. Country A has committed to international agreements that prioritize the procurement of the most advanced and effective vaccines available.

Which of the following statements, if true, would most strongly persuade the Ministry of Health in Country A to not renew their contract with Zenith Pharmaceuticals and instead award it to NovaGen Biotech?

- (a) Only II, III, and V
- (b) Only I, II, and V
- (c) Only I, IV, and V
- (d) Only III and IV
- (e) Only II and III

- 3. Directions:** In the question given below, a passage is given followed by five statements. Read the information carefully and answer the question that follows.

EcoLife Industries, a leading manufacturer of eco-friendly household products in Country M, has recently faced a significant decline in sales and market share due to the emergence of new competitors offering similar products at lower prices. Additionally, there have been allegations that EcoLife's products are not as environmentally friendly as advertised, leading to negative publicity and consumer distrust. The management is considering various courses of action to regain market position, restore consumer confidence, and ensure the company's sustainability.

Statements:

- I. Conduct an independent audit of the company's products to verify their environmental claims and publicly release the findings.
- II. Reduce the prices of all products to match competitors, even if it means compromising on quality.
- III. Invest in research and development to innovate new, genuinely eco-friendly products that meet higher environmental standards.
- IV. Launch an aggressive marketing campaign highlighting the company's commitment to the environment and discrediting competitors' products.
- V. Collaborate with environmental organizations to develop certification programs that validate the eco-friendliness of products in the market.

Which of the following courses of action should EcoLife Industries implement to effectively address the decline in sales, restore consumer trust, and ensure long-term sustainability?

- (a) Only I, III, and V
- (b) Only II and IV
- (c) Only I, II, and IV
- (d) Only III and V
- (e) Only II, III, and V

Directions (4-8): Read the following information carefully and answer the questions that follow.

There is a matrix that contains four rows (I, II, III, and IV) from top to bottom and four columns (I, II, III, and IV) from left to right. Words will be formed when we read left to right in a row and top to bottom in a column.

Conditions:

- I. No letter is repeated in any row or column.
- II. If a letter appears in a particular row or column, it should not appear again in that row or column.
- III. Each row must form a meaningful English word of four letters.
- IV. Alphabets are coded as A = 1, B = 2, C = 3, ..., Z = 26.
- V. The sum of the positional values of the letters in any row or column must be an even number.
- VI. Some letters are given at specific positions in the matrix.

Row I, Column II contains 'A'.

Row II, Column IV contains 'E'.

Row III, Column IV contains 'D'.

Row IV, Column I contains 'M'.

4. Which letter will come in Row I, Column I?
 (a) T (b) S (c) C
 (d) B (e) R
5. What is the sum of the positional values of the letters in Row II?
 (a) 36 (b) 40 (c) 44
 (d) 48 (e) 32
6. Which of the following words can be formed in Row III?
 (a) BATS (b) LINE (c) CORD
 (d) MELT (e) None of these
7. What is the difference between the sum of positional values of Row III and Column II?
 (a) 4 (b) 2 (c) 6
 (d) 10 (e) 8
8. Which letter will come in Row IV, Column IV?
 (a) T (b) S (c) E
 (d) N (e) O

9. **Directions:** In the question given below, a passage is given followed by five statements. Read the information carefully and answer the question that follows.

The government of Country X is planning to introduce a new agricultural policy called "AgriBoost" to enhance food security and boost farmers' income. The policy includes subsidies for high-yield genetically modified (GM) seeds, investments in advanced irrigation systems, and access to low-interest loans for farmers. The government asserts that AgriBoost will modernize agriculture, increase crop productivity, and ensure sustainable farming practices across the country.

Statements:

- I. Several environmental organizations have warned that the introduction of genetically modified seeds may lead to a loss of biodiversity and could harm native plant species.
- II. Studies have shown that advanced irrigation systems proposed under AgriBoost are not suitable for the varied topography and soil conditions prevalent in many regions of Country X.
- III. A significant portion of the farming community lacks awareness and training on the use of GM seeds and advanced irrigation technologies, leading to potential misuse and crop failures.
- IV. Countries that have implemented similar agricultural modernization policies have witnessed a substantial increase in crop yields and farmers' incomes over the past decade.
- V. The low-interest loans provided under AgriBoost have stringent eligibility criteria, which most small-scale farmers in Country X cannot meet.

Which of the following statements, if true, would most seriously weaken the government's plan to implement the AgriBoost policy?

- (a) Only I and II (b) Only II, III, and V
 (c) Only I, III, and V (d) Only IV
 (e) Only II and IV

10. **Directions:** In the question given below, a passage is provided followed by five statements. Read the information carefully and answer the question that follows.

The Ministry of Health in Country Y is planning to launch a nationwide program called "FitnessFirst" to combat the rising rates of obesity and lifestyle-related diseases. The program includes building public fitness centers in urban and rural areas, offering free health check-ups, and providing subsidized fitness classes. The government believes that by making fitness facilities more accessible, the population will adopt healthier lifestyles, reducing the burden of healthcare costs in the long term.

Statements:

- I. A significant portion of the population in Country Y lives below the poverty line and is more concerned with meeting basic needs like food and shelter than participating in fitness programs.
- II. Private gyms and fitness centers are offering advanced facilities and personalized training, attracting those who can afford them and leaving public centers underutilized.
- III. There is a cultural stigma in certain regions of Country Y against women participating in public fitness activities, which may limit the program's effectiveness.
- IV. Similar programs in neighboring countries have successfully reduced obesity rates and improved public health over the past five years.
- V. The government's budget allocation for the FitnessFirst program is limited, leading to concerns about the quality and maintenance of the public fitness centers.

Which of the following statements, if true, most weakens the Ministry of Health's decision to launch the FitnessFirst program?

- (a) Only I (b) Only II
 (c) Only III (d) Only IV
 (e) Only V

Direction (11-15): Read the following information carefully and answer the given questions.

When a word and a number rearrangement machine is given an input line of words and numbers, it arranges them following a rule. The following is an illustration of input and rearrangement.

Input: 7621 BRAVE 3845 TOUCH 6482 DREAM 5173 SHINE 8274 GLOW

Step I: 49730 AEBRV 9954 OUCHT 36591 AEDMR 25282 EIHSN 64383 OGLW

Step II: 4379 BFAQU 4599 PVBGS 61359 BFCLQ 22258 FJGMR 46833 PFKV

Step III: 419 BFQ 423 PVBGS 618 BFCLQ 613 FJGMR 186 PFKV

Step IV: 14 BFQ 9 BGPSV 15 BCFLQ 10 FGJMR 15 FPKV

Step V: 17 14 20 15 11

Step V is the last step for the above input. Find out the appropriate steps for the following input.

Input: 8536 GRACE 2947 MOVIE 6183 BLEND 7312 SMILE 4765 POWER

11. What is the third number from the left in Step II?
 (a) 24419 (b) 44665
 (c) 22639 (d) 4659
 (e) 46817
12. Which word is immediately to the right of 'FJPL' in Step III?
 (a) CVD (b) FJKLR
 (c) FEGP (d) FCMK
 (e) None of these
13. What is the sum of all the numbers in Step V?
 (a) 27 (b) 28 (c) 29
 (d) 30 (e) 31
14. Which of the following is the fourth word from the left in Step IV?
 (a) FJKLR (b) FPQV (c) CFKM
 (d) EJOT (e) DVIP
15. What is the difference between the highest and lowest number obtained in Step V?
 (a) 11 (b) 13 (c) 6
 (d) 5 (e) 0

Direction (16-19): Read the following information carefully and answer the given questions.

A certain number of persons are sitting in a row. Some of them are facing north and some of them are facing south. Each of them is from a different city. The person from Canada sits third to the right of Quincy who is facing south. Six persons sit between Hayden and the person who is from Canada. Hayden and the person from Canada are facing opposite directions. Morgon sits second to the left of Hayden. The person from Denmark sits second to the right of Morgon and facing the opposite direction to Morgon. Morgon sits sixth to the left of the one from Egypt. Tyler is an immediate neighbor of the one who sits fourth to the right of the one who is from Denmark and Daisy. The one who is from Egypt sits third to the right of the one who is from Canada. The one who sits fourth to the left of Wren sits at one of the extreme ends. The number of people sitting to the right of Wren is thrice the number of people sitting to the left of Wren. Immediate neighbors of Tyler are facing the same direction but opposite to Tyler. The one who sits fourth to the left of Daisy sits second to the left of the one who is seventh from one end. The number of persons sitting between Daisy and the one from France is thrice the number of people sitting between Wren and the one from France. Quincy is sitting at an extreme end. Wren and the one from France are facing opposite directions to Daisy.

16. The same number of persons are sitting to the left of Wren as to the right of _____.
 - (a) the one who is sitting second to the right of Hayden.
 - (b) the one who belong to Canada
 - (c) the one who belong to Egypt.
 - (d) The one who is sitting second to the left of Hayden
 - (e) the one who belongs to France.
17. Who is sitting fourth to the right of the person who is sitting second to the right of Morgon?
 - (a) the one who belong to Denmark
 - (b) the one who is sitting immediately right of the one who belong to Canada.
 - (c) the person who is sitting exactly in the middle of Hayden and Daisy.
 - (d) the person who is second to the right of the person who belongs to Egypt.
 - (e) the one who is sitting second to the left of Daisy.
18. Which of the following pair is sitting second to the right of each other?
 - (a) Tyler and the one who is from Canada
 - (b) Quincy and the one who is from Denmark
 - (c) The one who is from Egypt and Wren
 - (d) Hayden and Daisy
 - (e) The one who is from Canada and France
19. How many persons are sitting between Daisy and Quincy?
 - (a) One
 - (b) Seven
 - (c) Five
 - (d) Ten
 - (e) Eight

Direction (20-23): Read the following information carefully and answer the given questions.

Eight students A, C, E, F, G, H, K, and L are related to each other born in 1953, 1957, 1963, 1982, 1985, 1988, 2015, and 2018 not necessarily in the same order. Age is calculated from 2024. E's spouse has a niece who is F. The difference between F and F's mother's age is a multiple of 10. The difference between A and A's mother-in-law's age is a multiple of

five. E's uncle's age was prime but he was not the eldest. The difference between C and C's son is a prime number less than twenty. The difference between E and E's spouse's age is double the difference between the E and K age. A has a sibling only. E has a son. E's uncle is not C but E's uncle's age is a prime number. K is not married to A. E has a father. There is an equal number of males and females. C has no siblings. G is the grandmother of E's only son. G is the grandmother of L. E's father has no sibling of his own and E's mother has no sister.

20. Which of the following age is a perfect square?
 - (a) the one who is born immediately after E.
 - (b) L
 - (c) the one who is born immediately after G
 - (d) H
 - (e) the one who is born immediately after K.
21. In which of the year C is born?
 - (a) 1957
 - (b) 1982
 - (c) 1963
 - (d) 1953
 - (e) 1985
22. How is K related to F?
 - (a) Mother
 - (b) Father
 - (c) Sister
 - (d) Daughter
 - (e) Brother-in-law
23. What is the sum of the age of E and K?
 - (a) 60 years
 - (b) 81 years
 - (c) 90 years
 - (d) 52 years
 - (e) 50 years

Direction (24-27): Read the following information carefully and answer the given questions.

There are ten buses A, B, C, D, E, F, G, H, J, and K which are at the base station. Four platforms P1, P2, P3, and P4 where the first bus can reach at 5 AM, 6 AM, 7 AM, and 8 AM respectively. Platform P1 is the nearest Platform and Platform P4 is the farthest. There must be a gap of one hour between the two buses reaching a particular station. The speed of the buses is considered to be the same. At most three buses can have a platform as its destination.

Bus C and D started from the base station and reached the nearest platform in a gap of 1 hour. C started its journey earlier than D. C reached the destination at 6 AM. Bus A stopped its journey at a platform which is two more than Bus D. Bus J reached the last platform. Bus H's destination is not the same as the destination of C, E and K. Destinations of F and G are the same. The destination of F and G is not the same as the destination of D and J. Bus B's destination is the same as the destination of K. There is only one more bus at the platform where Bus J reached its destination. Bus E's final destination is not P1 and P4. Bus B's destination is ahead of Bus E's destination.

24. Which of the following platforms have more than two buses?
 - (a) P1 and P4
 - (b) P2 and P3
 - (c) P1 and P2
 - (d) P3 and P4
 - (e) P2 and P4
25. Which of the following statements is/are false?
 - I. Bus B's destination is P4.
 - II. Bus E and Bus F's destinations are the same.
 - III. Bus D's destination is ahead of Bus F's destination
 - (a) Only I and II
 - (b) Only II and III
 - (c) All I, II and III
 - (d) Only II
 - (e) Only I and III

26. Which of the following buses have destination as P3?
 (a) Bus A, Bus B, Bus K. (b) Bus F and Bus C
 (c) Bus A, Bus H (d) Bus J and Bus D
 (e) Bus H and Bus F
27. At what time Bus D reached the platform P1?
 (a) 9AM (b) 8AM (c) 7AM
 (d) 10 AM (e) 12 AM

Direction (28-31): Read the following information carefully and answer the given questions.

Ten employees Ava, Brooke, Charles, Daniel, Emma, Fraklin, Gloria, Harper, Ivan, and Jacob sit at two rectangular tables X and Y. Three of them like action novels, three like drama novels and the remaining like comedy novels. Four employees sit at table X and the remaining sit at table Y. The person sitting in the middle faces outside the center and the person sitting in the corner faces inside the center. People sitting at Table X sit in the middle of the table and at Table Y four persons sit in the middle and four sit in the corner. Two seats are vacant in Table Y.

Daniel sits at table X. Jacob does not sit at table Y. Gloria sits immediately left of Jacob and neither of them like drama. The one who likes comedy does not sit in the middle of table Y. Daniel likes comedy and sits opposite Gloria. Emma likes comedy and sits at table Y. Both the immediate neighbors of Emma like drama. Ava sits immediately left of the one who likes drama. Fraklin sits third to the right of Ava. Charles sits at table X. Brooke does not like drama and comedy. Two persons sit between Daniel and the one who likes drama. Ivan sits 2nd to the right of a vacant seat. Both the vacant seats are not adjacent to each other. One of the persons who likes action sits the immediate right of the one who likes comedy.

After the arrangement:

- Ivan, Jacob and Harper change their novels.
 - Fraklin and Gloria change their novels.
 - Daniel changes his novel with the one who sits immediately left of Ava.
 - Brooke and Emma change their place but not the novels.
- Note:** The final arrangement is obtained after the I, II, III, and IV changes are done.
28. Which of the following statements is/are false?
 I. Ivan, Gloria likes Action novels.
 II. Emma, Fraklin, and Jacob are sitting on the table Y.
 III. Ava and Ivan are immediate neighbors.
 (a) Only II (b) Only Ivan
 (c) Only III (d) Only Ivan and III
 (e) Only II and III
29. The person who is sitting third to the right of person who is sitting immediately right of Fraklin likes _____ novels.
 (a) Drama (b) Either Action or Comedy
 (c) Action (d) Comedy
 (e) Either Drama or Action
30. Who are the immediate neighbors of Jacob?
 (a) Emma and Harper (b) Gloria and Daniel
 (c) Ava and Brooke (d) Charles and Fraklin
 (e) Ava and Fraklin
31. Four of the following five are alike in a certain way and hence form a group. Which among the following one does not belong to the group?
 (a) Emma (b) Fraklin (c) Harper
 (d) Ivan (e) Ava

Direction (32-35): Read the following information carefully and answer the questions given below.

Eight persons A, B, C, D, E, F, G and H are born in four different years viz., 1964, 1977, 1982 and 1991 and also in two different months viz., June and July but not necessary in the same order. All they are wearing eight different watches namely Noise, Omega, Citizen, Casio, Tissot, Timex, Titan and Amazfit. No two persons born in same month of a year. Age of the persons is calculated as on December 2020. Each year two persons were born in two different months. Months are not considered for age calculation. The person who wears Noise born immediately before F. F was born in the month which has 30 days. The sum of the ages of B and the one who wears Omega is equal to perfect square of a number. There are only three persons born between E and the one who wears Casio. D was not born in odd numbered year. More than two persons born between H and one who wears Tissot. As many persons born between A and the one who wears Timex is same as the persons born after the one who wears Citizen. There are two persons born between D and the one who wears Titan. Age of the one who wears Casio is not a prime number but born in one of the months before the one who wears Citizen. There are only two persons born between the one who wears Casio and the one who wears Omega. Only one person born between B and the one who wears Titan, who was not born in the year 1964. C and the one who wears Tissot are born in different month. A was born immediately after the F. H does not like Timex. F was not born immediately before or after the one who wears Citizen.

32. Who among the following is the Youngest?
 (a) The one who wears Amazfit
 (b) C
 (c) The one who born immediately after F
 (d) A
 (e) Both (a) and (b)
33. The one who wears Noise born in which month and year?
 (a) June, 1977 (b) July, 1982
 (c) June, 1982 (d) July, 1964
 (e) None of these
34. Four of the following alike in a certain way and thus form a group. Which of the following one that does not belong to the group?
 (a) D (b) F (c) H
 (d) C (e) G
35. If 'H' is born between 'B' and 'C', then what can be the age of 'H'?
 (a) 33 years (b) 56 years
 (c) 29 years (d) 38 years
 (e) None of these

Direction (36-39): Study the following information carefully to answer the given questions:

Certain number of persons are standing in a row facing north. Each person has a carton which are kept above one another. Each carton is of different colour i.e. Maroon, Indigo, Crimson, Violet, Orange, Black, Blue and Yellow but not necessarily in the same order. Only three people sit between P and Q. W has carton of Blue colour and is kept at the bottom of the stack. Only four people sit between Q and R. S sits third to the right of T. Only six people sit between Q and U. Not more than three people sit between R and U. More than four people are between S and U. V sits third to the right of U. Only three cartons are kept between carton of S and carton of V. Carton of V is placed above carton of S. None of them sits between V and T. W sits eight to the left of R. The carton of Yellow colour is kept immediately above carton of V. Not more than three persons sit between P and W. Only three cartons are kept between

the carton of Violet colour and the carton of Maroon colour. The Indigo carton is immediately above the Black carton. Only one carton is kept between the carton of Yellow colour and carton of P. Carton of T is of Yellow colour. Only one carton is kept between carton of Q and the carton of Orange colour. Carton of V is not of Indigo colour. Carton of U is kept immediately below the carton of Orange colour. The carton of Violet colour is kept immediately above carton of T.

36. How many persons sit in the row?
 (a) Sixteen (b) Seventeen (c) Eighteen
 (d) Nineteen (e) Twenty
37. Carton of R is of which colour?
 (a) Crimson (b) Maroon (c) Violet
 (d) Black (e) Blue
38. How many persons sit between R and S?
 (a) Six (b) Seven (c) Eight
 (d) Nine (e) Ten
39. How many cartons are there between carton of Crimson colour and carton of S?
 (a) One (b) Two (c) Three
 (d) Four (e) Five
40. Each of the questions below consists of a question and three statements numbered I, II and III given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read the question and all three statements and give answer accordingly.
 Six employees E, F, G, H, I, J attend office on six different days starting from Monday to Saturday and each of them have their office on different floors viz. 1st, 4th, 5th, 6th, 8th, 9th. The one who has office on 1st floor attend office on which of the following day?
- Only two persons attend office in between G and H. The one who attend office on Wednesday has office on 8th floor. F has office on an even numbered floor. E attends the office immediately before I. G attends office before H and has office on an odd numbered floor.
 - No one attend office after J. Only three persons attend office in between the one who has office on 1st floor and the one who has office on 5th floor. The one who attend office immediately before I has office 1st floor.
 - G has office neither on 1st nor on 5th floor. F attends office immediately after H. The one who attend office on Thursday does not has office on 4th floor.
- If the data in statement I and II together are sufficient to answer the question, while the data in statement III are not required to answer the question.
 - If the data in statement I and III together are sufficient to answer the question, while the data in statement II are not required to answer the question.
 - If the data in statement II and III are sufficient to answer the question, while the data in statement I are not required to answer the question.
 - If the data in all three statements I, II and III together are necessary to answer the question.
 - If the data in all the statements, I, II and III even together are not sufficient to answer the question.
41. The question given below consists of two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read all the statements and give answer.
 Certain number of people are sitting in a row facing north. T sits fourth from the left end. Only one person sits between F and T. How many persons sit in the row?
- S sits fourth to the right of P. Less than three persons sit to the right of S. At least five persons sit between P and the person

sitting at the extreme right end. Three persons sit between F and P. At least two persons sit between P and T.

- S is the neighbour of G. There are at least 13 persons in the row. P sits fourth to the right of F. Only two persons sit to the right of S. Only two persons sit between P and G, who is not the neighbour of F. Either one or five persons are there to the left of F.
- Neither Statement I nor Statement II is sufficient to answer the question.
 - Only Statement II is sufficient to answer the question.
 - Both Statements I and II are sufficient to answer the question.
 - Only Statement I is sufficient to answer the question.
 - Either Statement I or Statement II is sufficient to answer the question.

Direction (42-45): Read the questions on the basis of the information given below.

There are twelve person – P, Q, R, S, T, U, V, W, X, Y, Z and M who live in a building with 5 floors (Numbered 1 to 5). There are 3 flats on each floor from left to right Numbered 1 to 3. Immediate right of flat 2 means flat 3. Flat above flat means same flat number of above floors. There are 3 flats empty in this building. There can be only 1 flat empty on any floor. Also same flat number cannot be empty on any of the floors i.e. if Flat number 1 is empty on floor number 1, then flat 1 of any other floor cannot be empty. They all have different weights. They all earn different amounts – ₹10,000, ₹20,000, ₹30,000, ₹40,000, ₹50,000, ₹60,000, ₹70,000, ₹80,000, ₹85,000, ₹90,000, ₹95,000, ₹1,00,000.

- Z lives on flat number 1 of floor number 3. One who earns ₹30,000 lives somewhere in the flat which is above Z's flat. One person lives between the one who earns ₹30,000 and V whose weight is 96 kg.
 - P earns ₹40,000 who lives just below V's flat. One of the flats on which P lives is empty but that is not immediate left of P's flat. The one whose weight is 76 kg earns ₹90,000 lives on the flat second to right of Z.
 - There are two floors between W whose weight is 72 kg and the one whose weight is 85 kg. W lives above the one whose weight is 85 kg. W does not earn ₹30,000. Y earns ₹95,000 and lives immediate right of the one whose weight is 85 kg. M and Y live on same floor. M earns ₹20,000 and his weight is 9 kg more than U.
 - There are two flats between M and an empty flat. U and the one who earns ₹50,000 lives on the same floor which is either floor number 1 or 2. One flat is empty on that floor. There is one empty flat between the one whose weight is 61 kg and the one who earns ₹10,000.
 - The one whose weight is 61 kg is above the one who earns ₹10,000. The one whose weight is 55 kg lives on floor number 1. U has got 25 kg less than S who does not live on top floor. R and empty flat are on same floor.
 - X lives on one of the flats on the floor on which V lives. The one who earns ₹1,00,000 lives immediate right of X. The one who earns ₹80,000 lives just above the one who earns ₹85,000. The one who earns ₹70,000 lives below the one who earns ₹85,000 such that no one lives between them. T and U lives on same floor. Z's weight is 15 kg more than P.
 - R lives on a floor on which one flat is empty. Weight of one of the person on same floor who is earning ₹85,000 is 50% of V. The one whose weight is 82 kg lives on one of the floors above the floor of one whose weight is 90 kg.
42. What is the total weight of V and the one who earns ₹10,000?
- 264 kg (b) 126 kg
 (c) 159 kg (d) 118
 (e) None of these

43. Who lives on flat number 2 of floor number 2?
 (a) S
 (b) The one who earns ₹60,000
 (c) The one whose weight is 90 kg
 (d) The one who earns ₹70,000
 (e) Cannot be determined
44. R lives on
 (a) Flat number 2 of floor number 2
 (b) Flat number 3 of floor number 3
 (c) Flat number 3 of floor number 1
 (d) Flat number 1 of floor number 4
 (e) None of these
45. How many floors are there between the floors of X and T?
 (a) Two (b) Three
 (c) One (d) None
 (e) Cannot be determined

ENGLISH LANGUAGE

Directions (46-50): In each of the questions given below three words are given in bold. These words may or may not be in their correct position. Following each sentence few sequences are provided. Select the sequence of the words which will make the given sentence contextually meaningful.

46. The scientist's **obfuscated** (A) on quantum physics was filled with technical terms that **presentation** (B) the topic for most of the audience, leaving them **bewildered** (C) and struggling to grasp the concepts.
 (a) BAC (b) ACB
 (c) BCA (d) Both (A) and (C)
 (e) All (A), (B) and (C)
47. The company's management team was divided into two **factions** (A), with one group **overtly** (B) advocating for a new product launch, while the other group **covertly** (C) worked to undermine the project behind the scenes.
 (a) BAC (b) ABC
 (c) ACB (d) Both (B) and (C)
 (e) All (A), (B) and (C)
48. The scientist, known for his **methodical** (A) and **deliberate** (B) research approach, embarked on a **ponderous** (C) study, carefully collecting and analyzing data over the course of several years to arrive at his groundbreaking discoveries.
 (a) ABC (b) BAC
 (c) CAB (d) Both (B) and (C)
 (e) All (A), (B) and (C)
49. As the young musician carefully **plucked** (A) the strings of his guitar, he **struck** (B) a chord that resonated through the room, **poking** (C) the audience's attention with the deliberate and melodious rhythm.
 (a) ABC (b) Both (A) and (C)
 (c) BAC (d) ACB
 (e) All (A), (B) and (C)
50. In the world of modern **technology** (A), the inner workings of cutting-edge quantum computers may seem **esoteric** (B) to most, but the smartphones we use daily have become utterly **ubiquitous** (C), altering the way we live and connect with the world.
 (a) BCA (b) Both (C) and (D)
 (c) ABC (d) ACB
 (e) All (A), (B) and (C)

Directions (51-55): Two paragraphs with a blank space are provided in each of the following questions. Fill in the blank space with the most appropriate one out of the four options given below, taking into consideration the consistency and compatibility of the two paragraphs. If none of the given options seems appropriate to fill in the blank space, choose option (e) as your answer choice.

51. I. Good afternoon, thank you for coming in today. Can you please let me know when you'd be available to start if you were to join our company and is there anything else you'd like to discuss or any questions you have about the company or the role?
 II. _____
 III. Certainly, our team is composed of highly skilled professionals who value collaboration. As for projects, we're currently working on several exciting initiatives, including a major product launch.
 (a) I'm ready to start as soon as possible, and there are no specific questions or concerns I have about the role or the company.
 (b) Yes, I do have a question. I'd like to know more about the team I'll be working with and the main projects I'd be involved in.
 (c) I'm curious about the work culture here. Can you describe the values and principles that define the company?
 (d) I'm interested in the overall work environment. Could you tell me more about the organization's welfare policies for its employees?
 (e) None of the above
52. I. In recent years, our country's economy has flourished, driven by strong domestic demand, strategic investments in key sectors, and a focus on innovation, resulting in a steady increase in our GDP and regional economic leadership.
 II. _____
 III. Despite significant economic progress, we acknowledge challenges like income inequality and sustainability. Our commitment is to build an inclusive, sustainable economy, sharing growth benefits and mitigating environmental impact, for a better quality of life for all.
 (a) Economic downturns can result from various factors, including financial crises, recessions, global economic issues, and overreliance on a single sector.
 (b) Our recent economic growth has been primarily driven by foreign investments, overshadowing the significance of domestic demand and innovation.
 (c) In contrast, our economic achievements have been primarily due to political stability, with minimal focus on technological advancements or diversification.
 (d) Economic success thrives on diversification, expanding beyond traditional sectors to invest in technology and services, enhancing resilience, and job market dynamism.
 (e) None of the above
53. I. Education forms the bedrock of individual and societal progress, providing the knowledge and competencies needed for personal development, economic well-being, and active citizenship, whether through formal or lifelong learning.
 II. Furthermore, education fosters critical thinking, problem-solving, and creativity. It empowers individuals to make informed decisions, contributes to economic competitiveness, and promotes social cohesion.
 III. _____
 (a) In today's globalized world, access to quality education is essential for reducing inequality and achieving sustainable development.
 (b) Education can perpetuate social inequalities by favoring privileged groups, and the pressure to conform to standardized testing can stifle creativity and critical thinking.

- (c) Despite its significance, education is solely focused on academic achievements, neglecting the development of essential life skills and practical knowledge.
- (d) Educational systems universally prioritize memorization over critical thinking, undermining the ability to adapt to real-world challenges and stifling creativity.
- (e) None of the above

54. I. _____

II. As the demand for freshwater resources rises due to population growth and industrialization, agriculture finds itself in a precarious position.

III. In the face of this challenge, the agricultural sector's resilience and productivity rely on the implementation of inventive solutions and sustainable practices.

- (a) Crops like rice, which require substantial amounts of water, are particularly vulnerable.
- (b) Water scarcity may prompt farmers to change their crop choices, favoring more drought-resistant varieties.
- (c) Water, the lifeblood of agriculture, has always played a pivotal role in the success of farming practices.

However, the world is facing a growing challenge: water scarcity

(d) Implementing water-efficient techniques like drip irrigation, mulching, and precision agriculture can significantly reduce water usage while maintaining crop productivity.

(e) None of the above

55. I. Baking, an age-old culinary art, has transcended its primary purpose of providing sustenance to become a source of joy, creativity, and a powerful stress-reliever for many.

II. _____

III. Baking brings happiness into our lives for a scientific reason. When we bake, our brains release endorphins, the "feel-good" chemicals that alleviate stress and boost joy.

- (a) In recent years, the popularity of communities marketing positive emotional and therapeutic aspects has surged.
- (b) Baking is often perceived as a melancholy process that only involves following precise instructions to achieve the desired results.
- (c) So, the next time you're feeling overwhelmed, why not head to the kitchen, don your apron, and immerse yourself in cooking?
- (d) The process of mixing flour, sugar, and eggs to create delicious treats not only satiates our taste buds but also brings happiness and nourishes our soul.

(e) None of the above

Directions (56-61): Read each sentence to find out whether there is any grammatical or idiomatic error in it. In the following sentences, a part is highlighted in each sentence which is error-free, the error, if any, will be in any other part of the sentence. The letter of that part is the answer. If there is 'No error', the answer is (e). (Ignore errors of punctuation, if any).

56. Scientists has discovered that environmental (A)/**factors play a greater role**/than genetic variants in determining (B)/ the height of children (C)/in low- and middle-income countries (LMICs). (D)

- (a) A (b) B (c) C
- (d) D (e) No error

57. Cybersecurity firms are turning (A)/to integrated platforms to offer (B)/**comprehensive online security for both/B2B** and end-users, as standalone tools (C)/struggle to meet enterprise security needs. (D)

- (a) A (b) B (c) C
- (d) D (e) No error

58. One of the student in the (A)/advanced mathematics class consistently (B)/**outperforms their peers**,/showcasing exceptional (C)/problem-solving skills.(D)

- (a) A (b) B (c) C
- (d) D (e) No error

59. Two studies published in (A)/**international journals have**/ found a worrying link among (B)/air pollution levels and (C)/the incidence of type 2 diabetes in Chennai and Delhi. (D)

- (a) A (b) B (c) C
- (d) D (e) No error

60. The global carbon (A)/**budget for a given temperature**/ limit is a global resource, common (B)/to the entire world, but (C)/is exhaustible and limited. (D)

- (a) A (b) B (c) C
- (d) D (e) No error

61. **The upgrade to the latest smartphone**/offers a bigger and better (A)/display, enhancing the user's (B)/experience for watching (C)/videos and browse websites. (D)

- (a) A (b) B (c) C
- (d) D (e) No error

Directions (62-67): In each of the following questions there are three blank spaces. Below each question there are some options and each option consist of three words which can be filled up in the blanks of the statement(s) to make the statement grammatically correct. Choose the correct alternative among the options given below.

62. The investigator, with _____ attention to detail, proceeded to _____ every piece of evidence, hoping to uncover the elusive truth behind the _____ case.

- (a) meticulous, scrutinize, complex
- (b) scandalous, ratify, complicated
- (c) trivial, foresee, veiled
- (d) virtuous, qualify, obvious
- (e) vivid, ridicule, intricate

63. In the face of the challenging situation, she took a _____ approach, focusing on _____ solutions rather than getting lost in _____ theories.

- (a) opportunistic, pragmatic, viable
- (b) delusional, durable, historic
- (c) pragmatic, practical, idealistic
- (d) feasible, realistic, implemented
- (e) obstruse, inevitable, probable

64. His _____ speech at the conference _____ the audience and _____ his deep knowledge on the subject.

- (a) vocal, intimidated, perplexed
- (b) exquisite, incapacitated, delivered
- (c) radical, repelled, illustrated
- (d) eloquent, captivated, conveyed
- (e) verbal, strengthen, intrigued

65. His _____ pursuit of the truth led to the _____ that _____ solved the long- standing mystery.

- (a) preserving, dimension, apologetically
- (b) tenacious, breakthrough, ultimately
- (c) conundrum, hesitation, eventually
- (d) indifferent, breach, gradually
- (e) apathetic, zenith, mistakenly

66. The _____ of how to allocate the _____ resources among competing projects required careful _____.

- (a) candor, restricted, analysis
- (b) procedure, inexhaustible, judgement

- (c) conundrum, limited, consideration
- (d) zealous, futuristic, admiration
- (e) mechanism, vicarious, precaution

67. The charity's mission to _____ poverty was a _____ endeavour, but their dedication never _____.
- (a) elevate, reckless, wandered
 - (b) prosecute, serendipity, capitulate
 - (c) prolific, sycophant, blurred
 - (d) reduce, malfeasance, distorted
 - (e) alleviate, quixotic, wavered

Directions (68-74): Read the following passage and answer the given questions.

A data breach may be analogously construed as an intramural foray transpiring under the shroud of nocturnal stillness, where an infiltrator, akin to a digital brigand, subtly _____ with invaluable assets, leaving the homeowner oblivious until the irrevocable damage manifests. In congruence, corporate entities find themselves ensnared in the surreptitious siphoning or compromise of consequential data within the digital milieu. This pervasive quandary casts its **ominous** pall across a myriad of global enterprises, with an astounding 83% of scrutinized organizations contending with recurrent incursions, as delineated in the comprehensive 2022 Cost of Data Breach analysis proffered by IBM.

Operationally, data breaches epitomize security conundrums culminating in the unauthorized manipulation, disclosure, access, or obliteration of personal data, opportunistically exploiting systemic susceptibilities or misconfigurations. Cyber malefactors, be they individual agents or collective entities, deploy an eclectic array of stratagems—ranging from malware incursions to the subterfuge of phishing emails—enabling their ingress into corporate networks. In an epoch characterized by the ceaseless generation and utilization of data across sundry devices, systems, and applications, the peril of clandestine data access is exponentially exacerbated.

The repercussions of data breaches transcend mere pecuniary ramifications, precipitating deleterious effects on an organization's reputational standing and instigating punitive measures from regulatory bodies. International jurisdictions, attuned to the burgeoning threat matrix, have responded by promulgating stringent data protection and privacy statutes, thereby endowing individuals with autonomy over their data and obligating corporate entities to assume augmented responsibilities. Consequently, organizational prioritization of preventative and responsive stratagems becomes imperative, cognizant of the exigency of compliance with dynamically evolving regulatory paradigms amidst the contemporaneous milieu of data proliferation and cyber-security vicissitudes.

68. What aspect contribute to the potential for unauthorized access?
- (a) Utilizing data security systems exclusively for the primary device.
 - (b) The continuous generation and utilization of data across diverse devices and applications.
 - (c) The continuous creation and technological advancement of devices.
 - (d) Collaborating with cyber malefactors, exposing all devices to vulnerabilities.
 - (e) Corporations compromising consequential data within the digital milieu.
69. Considering the prevalence of data across diverse domains, what are the potential consequences of failing to mitigate the risk of unauthorized access for businesses?

- I. Businesses may suffer reputational damage due to breaches of trust and the perception that they cannot adequately protect sensitive data
 - II. Financial losses through theft, fraud, or the compromise of sensitive financial information.
 - III. A diminution in customer satisfaction resulting from the dissemination of data pertaining to a company's product.
- (a) Only (I) (b) Only (II)
 - (c) Both (I) and (II) (d) Both (I) and (III)
 - (e) All (I), (II) and (III)

70. Which of the following cannot be considered as a data breach in the realm of cyber threats?

- (a) Phishing attacks that trick individuals into revealing sensitive information.
- (b) Malware infections exploiting vulnerabilities to compromise computer systems.
- (c) Utilizing standard network monitoring tools to detect and address typical network issues.
- (d) Insider threats involving malicious actions or negligence from within an organization.
- (e) None of these

71. How do businesses navigate compliance with stringent data protection regulations?

- (a) Assessing and ensuring the data protection regulations formulated by their cartel.
- (b) Designating a Data Protection Officer (DPO) can facilitate compliance efforts
- (c) Conducting a thorough data mapping and inventory helps identify what personal data is collected and processed
- (d) Both (a) and (c)
- (e) The passage does not provide enough information

72. Choose the most suitable word to fill in the blank.

- (a) elucidates (b) absconds
- (c) vilifies (d) exacerbates
- (e) ruptured

73. What is the tone of the passage?

- (a) Pessimistic (b) Informative
- (c) Sarcastic (d) Didactic
- (e) Analytical

74. According to the passage, choose the most suitable word that has a similar meaning as "Ominous".

- (a) propitious (b) threatening
- (c) cacophony (d) ubiquitous
- (e) disparate

Directions (75-80): Read the following passage and answer the questions.

Voice or speaker recognition refers to a machine or program's ability to interpret dictation or understand and execute spoken commands. This technology has become increasingly prominent with the _____ of artificial intelligence (AI) and intelligent assistants like Amazon's Alexa and Apple's Siri. Offering hands-free interactions, voice recognition systems enable users to make requests, set reminders, and perform various tasks simply by speaking.

The process involves automatic speech recognition (ASR) programs that can identify and differentiate voices. Some ASR programs **necessitate** users to train the system for **improved** accuracy in converting speech to text. Evaluation of a voice's frequency, accent, and speech flow is **integral** to voice recognition systems.

While voice recognition and speech recognition are often used interchangeably, they have distinct meanings. Voice recognition identifies the speaker, whereas speech recognition assesses the content of what is said.

In practice, voice recognition software on computers converts analog audio into digital signals through analog-to-digital (A/D) conversion. This digital database of words or syllables is then compared to signals during pattern recognition. The size of the program's effective vocabulary depends on the computer's RAM capacity, with faster processing speeds enhancing search capabilities.

Voice recognition involves analyzing speech through models like the hidden Markov model or recurrent neural networks. The former breaks down spoken words into phonemes, while the latter uses previous outputs to influence current inputs, improving capabilities and accuracy with increased data.

The integration of voice recognition into smartphones and home devices like Google Home and Amazon Echo has made this technology ubiquitous. As more users engage with voice recognition, the wealth of data generated enhances the capabilities and accuracy of these systems, indicating a promising trajectory for the technology's future.

75. What is the specific focus of voice recognition in comparison to speech recognition?

- (a) Voice recognition predominantly centers on discerning the speaker's identity rather than appraising the uttered content.
- (b) Voice recognition deciphers the speaker's words and then correlates them with nearby speakers.
- (c) Speech recognition primarily concentrates on the speaker's tone, whereas voice recognition identifies the speaker's voice modulations.
- (d) The terms voice recognition and speech recognition are frequently used interchangeably, as they serve a similar purpose.
- (e) Voice recognition entails discerning the distinct acoustic modulations of the speaker, whereas speech recognition proffers lexical conjectures to the speaker.

76. What is the function of pattern recognition as described in the passage in the context of voice recognition software?

- (a) Translating analog audio into digital signals
- (b) Recognizing the digital signals and subsequently converting them into analog signals for the system to comprehend the language.
- (c) Analyzing and comparing a digital repository of words derived from converting analog audio into digital signals
- (d) Evaluating the computer's RAM capacity to expedite the search for an effective vocabulary.
- (e) Both (a) and (c)

77. How does Markov model contribute to the overall improvement of voice recognition capabilities?

- (a) Incorporating and neglecting the preceding output to impact the input of the subsequent stage in voice recognition.
- (b) Utilizing phonemes for coarse identification of the origin of words and their association with the correct meaning.
- (c) Aggregating words into components to generate the output for the voice recognition process.
- (d) Dissection of words into phonetic components allows for a more granular analysis of voice recognition.
- (e) None of these

78. Why does the passage suggest that the wealth of data is a crucial factor in determining the positive course of development for voice recognition technology?

- (a) The substantial amount of data will allow users to obtain authentic results of their search requests.
- (b) It will provide a diverse and extensive range of speech patterns, accents, and variations, enriching the system's database.
- (c) More users will be engaged to contribute to the wealth of data
- (d) It enhances the capabilities of voice recognition systems to accurately interpret and respond to a wide array of voices and linguistic nuances
- (e) Both (b) and (d)

79. Choose the most suitable word that should fill the given blank.

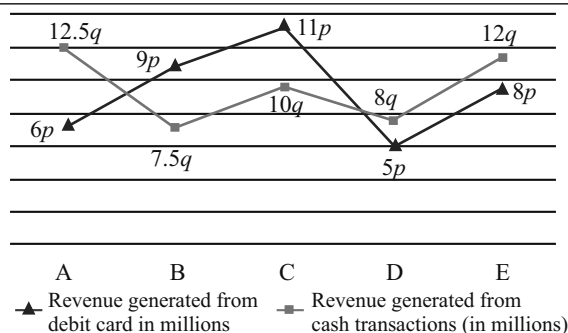
- (a) advent
- (b) oblivious
- (c) conservation
- (d) adversity
- (e) mundane

80. Choose the most appropriate choice that can substitute the three highlighted words in the passage in the same order.

- (a) require, alleviated, vital
- (b) rationalize, progressed, exclusive
- (c) crucial, discern, inherent
- (d) mandate, elevated, indispensable
- (e) impose, augmented, resilient

DATA ANALYSIS AND INTERPRETATION

Direction (81-84): Answer the questions based on the information given below. The given line graph shows the revenue generated from debit card transactions and cash transactions for five different stores, A, B, C, D and E.



Note:

- (a) If the total revenue generated worldwide (from debit card and cash transactions) is less than 400 million, then the store is given a '3' rating.
- (b) If the total revenue generated worldwide (from debit card and cash transactions) is between 400 million and 500 million, then the store is given a '2' rating.
- (c) If the total revenue generated worldwide (from debit card and cash transactions) is more than 500 million, then the store is given a '1' rating.
- (d) The sum of revenue from debit card transactions for stores 'A' and 'E' is 420 million, and the difference between the revenue from cash transactions for stores 'C' and 'D' is 48 million.

81. Find the ratio of the total revenue generated from store 'A' and 'B' together from debit card transactions to the total revenue generated from store 'B' worldwide.

- (a) 2 : 3
- (b) 1 : 2
- (c) 1 : 1
- (d) 3 : 4
- (e) None of these

82. For the store that received a '3' rating, out of the total revenue generated from debit card transaction, 60% was paid by males, and the rest was paid by females. Meanwhile, out of the total revenue generated from cash transactions, 25% was paid by males, and the rest was paid by females. Find the revenue generated worldwide from females for the store that received a '3' rating.
- (a) 252 million (b) 104 million
(c) 180 million (d) 204 million
(e) None of these
83. Which of the given store is/are given the rating '2'?
- (a) Store A (b) Store B
(c) Store B and C (d) Store D
(e) Store A and B
84. How much percent more is the total revenue generated worldwide for the stores that received a '1' rating compared to the revenue generated from cash transactions for store A?
- (a) 266% (b) 220%
(c) 325% (d) 100%
(e) None of these

Direction (85-87): Study the following data carefully and answer the questions accordingly.

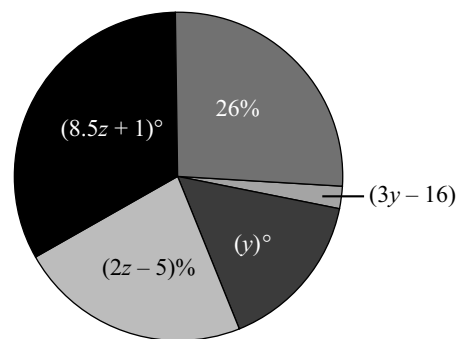
The given table shows data regarding the total number of vehicles (cars + motorcycles) in five different cities. It also displays the percentage/number of cars and the percentage of motorcycles in each city relative to the total number of vehicles in that city. Some of the data is missing from the table. You need to calculate the missing data and answer the questions based on it.

Cities	Total number of vehicles in the cities	Percentage/number of cars in the city	Percentage of motorcycles in the city
Pune	—	480	—
Delhi	—	600	25%
Jaipur	—	30%	—
Punjab	900	—	—
Haryana	—	—	60%

85. Let 'a' represent the difference between the number of cars and motorcycles in the city of 'Jaipur,' and 'b' represent the total number of vehicles in the city of 'Jaipur.' Which of the following options accurately expresses the relationship between 'a' and 'b'?
- (a) $5a = 7b$ (b) $4a = 3b$
(c) $5a - 3b = 0$ (d) $5a - 2b = 0$
(e) $5a = 2b$
86. Suppose the number of vehicles in the city of 'Punjab' is 25% less than the number of vehicles in the city of 'Pune', what is the percentage of motorcycles among the total number of vehicles in Pune?
- (a) 39% (b) 72% (c) 60%
(d) 65% (e) None of these
87. Suppose the ratio of the number of cars in cities 'Jaipur' and 'Punjab' is 5 : 9, and the difference between the number of motorcycles in city 'Jaipur' and 'Punjab' is 340, what is the total number of vehicles in city 'Jaipur'?
- (a) 1600 (b) 1080 (c) 1500
(d) 1000 (e) None of these

Direction (88-91): Study the following information carefully and answer the questions.

The following pie chart shows the distribution of the number of laptops sold by five stores.



● Store A ● Store B ● Store C ● Store D ● Store E

Conditions:

- I. The total number of laptops sold by all the five stores (A, B, C, D and E) taken together is 7200.
II. $(19z - 4y) = 42$
III. $z^2 + 2z - 224 = 0$

Note: The values given in the above given pie chart in the form of degree or percentage form cannot be negative.

88. If 38% of the all laptops sold by store E are gaming laptops and remaining are non gaming laptops, then find out the number of non gaming laptops sold by store E.
- (a) 1308 (b) 1372 (c) 1250
(d) 1544 (e) 1488
89. The average number of laptops sold by store A, C, E and B together is how much more than the average number of laptops sold by store B and D together?
- (a) 504 (b) 397 (c) 518
(d) 482 (e) None of the above
90. If the difference between the number of laptops sold by store D and E is 75% less than the number of laptops sold by store F and the ratio between the number of laptops sold by store F and the number of laptops unsold by the same store is 62 : 75, then find out the number of laptops unsold by store F.
- (a) 4800 (b) 3600 (c) 4000
(d) 3900 (e) 3500
91. If the MRP of each laptop of store A is ₹68000 and $\left(\frac{5}{6}\right)$ of the total number of laptops sold by store A were sold on MRP and remaining were sold at 'a'% discount, then find out the value of 'a'. It is assumed that the total amount collected by store A by selling the laptops is ₹122204160.
- (a) 28 (b) 40 (c) 21
(d) 33 (e) 24
92. Suppose five friends Ritu, Mohani, Aman, Sam and Kiran are appointed to complete a certain piece of work. Ram and Mohani together can complete a work in 24 days. Mohani and Aman together can complete 50% of the work in 15 days. Aman and Sam together can complete the work in 24 days. Ritu and Kiran together can complete 16.66% of the work in 6 days. Sam and Kiran together can complete the work in 20 days.
- Suppose Bittu is 50% more efficient than Aman then, what will be the time taken by Bittu alone to complete 40% of the work.
- (a) 54 days (b) 48 days (c) 36 days
(d) 15 days (e) None of these
93. 'S' undertook an investment endeavor by investing ₹ $a + 1000$ into a scheme yielding a simple interest rate of 20% per annum for a duration of three years. Simultaneously, 'A' ventured into a financial enterprise by investing ₹ $a - 4000$ in a compound interest scheme at a 20% annual rate for a three-year period. If the difference between interest earned from both investments is ₹1,080, then which of the following statement(s) is/are true?

- I. The variable 'a' can take multiple values.
 II. The minimal value for 'a' is less than 10,000.
 III. 'a' is an even integer.
- (a) Only I (b) Only I and II
 (c) Only III (d) Only I and III
 (e) None of these

94. Three friends, 'X', 'Y', and 'Z', decided to start a business together. They made investments of ₹6,000, ₹8,000, and ₹5,000, respectively. The time for which 'X', 'Y', and 'Z' kept their money invested is 't' months, (t - 3) months, and (t + 3) months, respectively. During the period for which 'X' had his investment, he also worked as a manager for which he receive a monthly salary of ₹3,000 from the gross profit. At the end of the business venture, the amount received by 'X' was ₹25,000 more than that received by 'Z'. The total profit was ₹81,000.

Find the profit share of Y.

- (a) ₹12,000 (b) ₹19,000 (c) ₹24,000
 (d) ₹25,000 (e) ₹16,000

95. In a plastic bag, there are (a + 5) red rubber bands, (2b - 5) blue rubber bands, and (2.5a - 5) black rubber bands. The number of red rubber bands is 40% less than the number of blue rubber bands, which is 25% more than the number of black rubber bands. In a separate box, there are (a + b) white rubber bands, (3b - a) orange rubber bands, and (b - a) yellow rubber bands. Calculate the probability of randomly selecting 2 white rubber bands, from the box, given that the selection is done without replacement.

- (a) $\frac{15}{104}$ (b) $\frac{4}{13}$ (c) $\frac{35}{104}$
 (d) $\frac{6}{17}$ (e) None of these

Direction (96-97): "You are presented with two quantities, labeled as Quantity I and Quantity II. Your task is to solve both quantities and determine the correct relationship between Quantity I and Quantity II. After solving, choose the appropriate option that describes the relationship correctly."

- (a) Quantity-I > Quantity-II
 (b) Quantity-I < Quantity-II
 (c) Quantity-I ≤ Quantity-II
 (d) Quantity-I = Quantity-II or No relation
 (e) Quantity-I ≥ Quantity-II
96. **Quantity-I:** The time required for tap 'P' to fill a water tank is 4 hours more than the time taken by tap 'Q' and tap 'Q' together to complete the same task. If tap 'P' takes 'p' hours to fill the tank independently, deduce the value of '25p,' given that tap 'Q' can independently fill the tank in 15 hours.
- Quantity-II:** Determine the number of distinct ways to partition a group of 7 individuals into three subgroups. One of these subgroups must consist of 3 members, while the remaining subgroups should each have 2 members."
97. **Quantity-I:** The monthly income of Tarun is ₹(z + 3300). He spent 'y%' on education and (1/3) of the remaining was spent on house rent. After that ₹652 was spent on travelling and (y + 14)% of the remaining was saved in bank and remaining was saved in the post office. If the amount spent on house rent by him is ₹12826 and (7y² - 476y + 8092 = 0), then find out the 2.5% of 'z'.
- Quantity-II:** If 1.2P² = 2268750, then find out the value of 'P'.

Direction (98-99): Study the following information carefully and answer the given question.

Observe the Series I carefully to identify the logic and obtain the value of 'P'. In both of the series, the same logic is applied.

Series I: 60, 68, 14, (P), -294, 786, -1272

Series II: (P-12), Q, R, S, T, U, W

98. What is the value of 'R'?

- (a) -6 (b) 152 (c) 148
 (d) 168 (e) 145

99. If the average of the values of 'S' and 'U' will be equal to the first term of a newly formed third series, then which of the following statements is/are true? It is assumed that each of the series is following the same logic in this question.

- (i) The sum of the values of 'T' and 'W' is the multiple of 9.
 (ii) The fourth term of the third series is an odd number.
 (iii) The difference between the value of 'Q' and the second term of the third series is 426.
- (a) Only (i) and (iii) (b) Only (ii)
 (c) Only (ii) and (iii) (d) Only (iii)
 (e) None is true

Directions (100-104): Read the table carefully and answer the following question.

The table shows the data about distance (km) covered from one point to another point by the vehicle used between each point and speed of each vehicle (km/h).

Note:

- (i) If one person wants to travel from P to S, then he/she can only use a vehicle available at the point P.
 (ii) ** means there is no direct connection between given points.

	P	Q	R	S	Transport Vehicle	Speed (km/h)
P	0	360	**	780	Train	60
Q	540	0	**	600	Taxi	50
R	420	720	0	**	Bus	40
S	630	**	900	0	Bike	45

100. If Vasu travels from P to S and Sanjeet travels from Q to P, then find the difference between the maximum distance travelled by both of them.
 (a) 245 (b) 260 (c) 250
 (d) 240 (e) None of these
101. If Gaurav wants to travel from P to R, then find the minimum time required by him to cover this distance.
 (a) 30 hours (b) 37.5 hours (c) 33 hours
 (d) 25 hours (e) 42 hours
102. If Hemant travels from P to S via Q and the fair of the vehicle per km from point Q to S is 25% more than that of from P to Q, then the total fair collected is ₹19980. Find the train fare per km.
 (a) ₹25 (b) ₹18 (c) ₹32
 (d) ₹40 (e) ₹30
103. Find the minimum possible sum of the time (in hours) taken to cover distance from P to S and from R to S.
 (a) 42.5 (b) 27.5 (c) 33.2
 (d) 25.4 (e) 36.5
104. A three-digit number PQR, when multiplied by a single digit number X, then the product is 2634. In the number PQR, the digit at tens place is equal to half of X. Which of the following statements is/are true about PQR?
 (i) Unit digit is 9
 (ii) It is a prime number
 (iii) Sum of the digits is 18
- (a) Only (i) and (iii) (b) Only (i) and (ii)
 (c) Only (iii) (d) All (i), (ii) and (iii)
 (e) None of these

Directions (105-106): Read the information carefully and answer the following questions.

A cube of side C cm is placed inside the sphere of radius R cm in such a way that sphere touches all the vertex of the cube. A cone of radius $\sqrt{3}R$ cm and height H cm has volume 3168 cm^3 .

105. What is the ratio of the total surface area of the sphere to the lateral surface area of the cube?

- (a) 33 : 17 (b) 33 : 19 (c) 31 : 17
(d) 33 : 14 (e) 29 : 17

106. What is the relation between the C and H?

- (a) $C^2 = \frac{1600}{H}$ (b) $C^2 = \frac{2300\sqrt{3}}{H}$
(c) $H^2 = \frac{1125\sqrt{3}}{C}$ (d) $C^2 = \frac{1200\sqrt{3}}{H}$
(e) $C^2 = \frac{1344}{H}$

Directions (107-108): Solve the number series to answer the following questions.

$\sqrt[3]{A}$, 31, B, C, 439, B + 687, 1296

Note:

- (i) All the terms in the given series and 'n' are positive integer.
(ii) Difference between 31 and B is X, which has 3 factors excluding X itself, those are 13, 5 and (A - 8).

107. What is the value of (A + C - n)?

- (a) 229 (b) 232 (c) 221
(d) 224 (e) 225

108. What is the LCM of (B + 34) and X?

- (a) 101 (b) 102 (c) 130
(d) 104 (e) 195

Directions (109-111): Solve the quadratic and answer the following questions.

$$A : (x-2)^2 = (-3x^2) + 2^2 + 25x - P$$

$$B : \left(10y^2 - 3^2y + \frac{2}{3}\right) \times 3 + 10y = 0$$

One root of equation A is 5.

109. $\frac{7P}{P-24} \times 0.2P - 91$ is equal to

- (a) P + 2 (b) 2P + 2 (c) 2P - 1
(d) P - 1 (e) None of these

110. Find the product of smallest root of equation A and smallest two-digit prime number.

- (a) $13\frac{1}{4}$ (b) $17\frac{1}{4}$ (c) $24\frac{3}{4}$
(d) $13\frac{3}{4}$ (e) $23\frac{3}{7}$

111. Which of the following are the smallest root of equation A and largest root of equation B-respectively?

- (a) $\frac{9}{4}, \frac{2}{5}$ (b) $\frac{2}{5}, 5$ (c) $\frac{9}{4}, \frac{1}{6}$
(d) $\frac{2}{3}, \frac{3}{5}$ (e) None of the above

$$(2a^3 + 2b^3)(a-b)$$

112. Quantity I: $\frac{(8a^2 - 9ab + 2b^2 + 7ab - 6a^2)(a^2 - b^2)}{(2a^3 + 2b^3)(a-b)}$

Quantity II: Value of P - 2Q if $Q < (-1) < P$, Where P and Q are integers.

- (a) Quantity I < Quantity II (b) Quantity I > Quantity II
(c) Quantity I = Quantity II (d) Quantity I Quantity II
(e) Quantity I Quantity II

113. A four-digit odd number 'M' whose unit digit is not 5 when divided by a prime number gives resultant as a three-digit number 'N' whose tens place digit is 0. The sum of digits of M is 8 while that of N is 5. If $M < 3400$ then which of the following statement/s is/are correct about M and N?

- I. Product of the digits of M is 3 more than the sum of the digits of N
II. Difference between the hundred place digits of M and N is 2
III. Sum of M and N is 1729
(a) Only (II) (b) Only (I) and (II)
(c) Only (II) and (III) (d) Only (I)
(e) All of the above

Directions (114-115): Read the information carefully and answer the questions.

A, B, C and D together can do a work 'X' in 14 days. The ratio of efficiency of A to that of C is 3 : 2. C is 50% less efficient than that of B and B is $33\frac{1}{3}\%$ more efficient than D. A and B together can complete

work 'X' in 'x' days, while A and D can do the same work together in 'y' days.

114. Two persons P and Q together can complete another work 'Y' in (x - 9) days, while P and R together can complete the same work in (y - 4) days. If P, Q and R together can complete the work 'Y' in

$\frac{3x}{0.5y-8}$ days, then efficiency of 'R' is what percent more or less

than the efficiency of 'Q'?

- (a) 80% (b) 120% (c) 100%
(d) 150% (e) 60%

115. 'j' men can complete a work in (x + 4) day, while '(y - 8)' men can complete the same work in 'k' days. If '(y + 2)' men can complete the same work in (0.5k + 3.5) days, then find in how many days

- (k - j) men can complete the same work.
(a) 21 days (b) 42 days (c) 70 days
(d) 64 days (e) 84 days

GENERAL, ECONOMY/BANKING AWARENESS

116. An 8-member high-level panel has been formed by the government to address the concept of "One nation, One election. Who is the chairman of the committee?

- (a) Rajnath Singh (b) Manmohan Singh
(c) Ramnath Kovind (d) Ravishankar Prasad
(e) None of the above

117. Which of the following organizations has joined One CGIAR integrated partnership?

- (a) ICRISAT (b) ICAR
(c) DRDO (d) ISRO
(e) None of the above

118. The cabinet recently granted approval for a viability gap funding of Rs. 3760 for a battery energy storage system. This initiative aims to facilitate the establishment of 4000 MW projects by the fiscal year 2030-2031, with financial assistance covering up to what percentage of the total capital cost?
- (a) 20% (b) 25% (c) 30%
(d) 40% (e) 50%
119. SIDBI inks MoU with India Post Payments Bank (IPPB) on October 11, 2023 for development and financing of _____ in rural and remote areas of the country by leveraging strengths of respective Institutions.
- (a) Informal micro enterprises
(b) MSMEs
(c) Formal Micro Financing Agencies
(d) Traditional artisans
(e) None of the above
120. Psyche is a NASA mission to study a metal-rich asteroid with the same name, located in the main asteroid belt between which planets?
- (a) Mercury and Mars (b) Mars and Jupiter
(c) Venus and Mars (d) Jupiter and Saturn
(e) None of the above
121. The Reserve Bank of India's Digital Payments Index (RBI-DPI), has increased across all parameters driven by significant growth in payment infrastructure and payment performance across the country over the period, which of the following parameters has the highest weight for RBI-Digital Payments Index?
- (a) Payment Enablers
(b) Payment Infrastructure Demand-side factors
(c) Payment Infrastructure Supply-side factors
(d) Payment Performance
(e) Consumer Behaviour
122. In October 2023, Dr. Joyeeta Gupta received the Spinoza Prize, the highest distinction in Dutch science for her work in the field of _____.
- (a) Climate Change (b) Electronic Waste
(c) Logarithmic Solution (d) Ocean acidification
(e) None of the above
123. Ashwini Vaishnav inaugurated India's first 3D printed post office building in Bengaluru. It has been constructed by _____ company along with _____.
- (a) L & T and IIT Madras (b) NTPC and IIT Kanpur
(c) Tata and IIT Delhi (d) Birla sons and IIT Bombay
(e) None of the above
124. Which fintech company has launched a new shopping app called "Pincode" which is built on top of the ONDC platform?
- (a) Paytm (b) PhonePe
(c) Google Pay (d) MobiKwick
(e) None of the above
125. In October 2023, the Reserve Bank of India (RBI) approved whom to acquire 9.99 percent of the paid-up share capital in private lender IndusInd Bank?
- (a) SBI mutual fund (b) LIC
(c) Barclays Bank PLC (d) BNP Paribas
(e) None of the above
126. Which of the following banks has launched the country's 1st ever numberless credit card with no CVV expiry date?
- (a) RBL Bank (b) Kotak Mahindra Bank
(c) Axis Bank (d) Yes Bank
(e) IDFC First Bank
127. Which of the following banks has introduced a facility to allow customers to enroll in social security schemes such as Pradhan Mantri Jeevan Jyoti Bima Yojana, the Pradhan Mantri Suraksha Bima Yojana, and the Atal Pension Yojana by just furnishing an Aadhaar card?
- (a) Bank of India (b) Bank of Baroda
(c) Union Bank of India (d) State Bank of India
(e) Canara Bank
128. The Reserve Bank of India (RBI) has announced that it will release _____ of the funds held under the Income-Recognition and Asset Classification (I-CRR) framework on September 9th, _____ further 25% will be released on September 23rd, and _____ on October 7th.
- (a) 15%, 35%, 50% (b) 50%, 25%, 20%
(c) 25%, 25%, 50% (d) 35%, 30%, 40%
(e) None of the above
129. Which of the following global initiatives is led by India?
- (a) Climate Action Network (CAN)
(b) International Solar Alliance (ISA)
(c) Coalition for Disaster Resilient Infrastructure (CDRI)
(d) World Health Organisation
(e) None of the above
130. According to the October 2023 World Economic Outlook report by the IMF, what will be India's GDP growth rate in the FY24 & FY25?
- (a) 6.1% (b) 6.3% (c) 6.5%
(d) 6.7% (e) 7.1%
131. The World Heritage committee has listed Sacred Ensembles of Hoysalas as the 42nd World heritage site in India. Which of the following are there in the Hoysalas temple list?
- (a) Channakeshava Temple (b) Keshava temple
(c) Hoysaleswara temple (d) Only (a) and (c)
(e) All (a), (b) and (c)
132. Recently Sikkim experienced a Glacial Lake _____ Flood in the teesta river basin over Lhonak lake killed seven people. What will come in place of blank space?
- (a) Outburst (b) Outwash
(c) Overflowed (d) Overthrust
(e) None of the above
133. What is the name of the Israel cutting edge rocket defense system?
- (a) HIMARS rocket (b) Iron dome
(c) Iron Dome (d) National missile defense
(e) None of the above
134. According to the Ministry of Petroleum and Natural Gas (MoPNG), it will add as much as 56.6 million tonnes per annum (mtpa) of crude oil refining capacity in the next seven years (by 2030). What is the rank of India in crude oil refining?
- (a) 3rd (b) 4th (c) 5th
(d) 6th (e) 7th
135. TRIFED Showcases India's Tribal Craftsmanship of Longpi pottery at G-20, Longpi pottery is of which state?
- (a) Arunachal Pradesh (b) Nagaland
(c) Assam (d) Manipur
(e) Mizoram
136. In September 2023, NASA astronaut Frank Rubio made history by returning to Earth after an extraordinary 371 days in space, shattering previous records. However, the ultimate record for the longest duration spent in space remains with the late Russian cosmonaut Valeri Polyakov, who completed an astounding _____ aboard Russia's Mir space station between January 1994 and March 1995.
- (a) 417 days (b) 427 days (c) 437 days
(d) 447 days (e) 457 days
137. The ferry service will run between the port town of Nagapattinam in Tamil Nadu and Kankesanthurai, a resort hub in the Jaffna district of northern Sri Lanka, covering a total distance of _____.
- (a) 60 Nautical miles (b) 65 Nautical miles
(c) 70 Nautical miles (d) 75 Nautical miles
(e) 90 Nautical miles

138. Who recommends the distribution of the net proceeds of taxes between the Union government and the state governments?
 (a) NITI Aayog (b) Yojana Aayog
 (c) Finance Commission (d) Ministry of Finance
 (e) None of the above
139. Which of the bank has partnered with Athletics Federation of India (AFI) to support women athletes of the country, especially the youngsters.
 (a) HSBC (b) Citi Bank
 (c) BNP Paribas (d) DBS Bank India Limited
 (e) Deutsche Bank
140. All the Nobel Prizes are awarded in Stockholm, Sweden. The Nobel peace prize is awarded in which country?
 (a) Oslo, Norway (b) Helsinki, Finland
 (c) Copenhagen, Denmark (d) Madrid, Spain
 (e) Brussels, Belgium
141. Who are the parties / stakeholders in a tokenisation transaction?
 (a) Merchant (b) Merchant's acquirer
 (c) Issuer (d) Customer
 (e) All of the above
142. Which of the following sports is not included in Olympic 2028?
 (a) Cycling (b) Billiards
 (c) Golf (d) Handball
 (e) Fencing
143. Which of the following is part of the primary market?
 (a) Public issue (b) Preference issue
 (c) Right offering issue (d) Bond issue
 (e) All of the above
144. The Central Board of Direct Taxes (CBDT) has issued a directive aimed at ensuring that Department for Promotion of Industry and Internal Trade (DPIIT) recognized start-ups are not burdened with unnecessary scrutiny under the revised _____ provisions in the Finance Act, 2023.
 (a) Corporate Tax (b) Service Tax
 (c) Angel Tax (d) Capital Gain Tax
 (e) Income Tax
145. The Valley Queen Heritage Train is a unique railway service in which of the following states?
 (a) Goa (b) Himachal Pradesh
 (c) Rajasthan (d) Tamil Nadu
 (e) Kerala
146. Which of the following is the owner of public general insurance companies in India?
 (a) IRDAI (b) LIC
 (c) Government of India (d) RBI
 (e) None of the above
147. Which of the following is not a theme of Enhanced Access and Service Excellence-5.0 (EASE-5.0)?
 (a) Business growth (b) Profitability
 (c) Risk (d) Customer Service
 (e) Know Your Customer
148. Direct Benefit Transfer scheme aims to transfer subsidies directly to the people through their linked bank accounts. DBT is implemented in which of the following scheme?
 (a) PM Awas Yojana
 (b) National Food Security Mission
 (c) Pradhan Mantri Krishi Sinchai Yojana
 (d) PM KISAN
 (e) All of the above
149. India and _____ became the first countries outside the US to get access to the general search engine experience via Google search.
 (a) China (b) Japan
 (c) Australia (d) Singapore
 (e) None of the above
150. Which of the following are included in top three broker firm in India?
 (a) Zerodha, Groww, Angel One
 (b) Upstox, Sharekhan, Motilal Oswal Financial Services Ltd
 (c) Axis Direct, Kotak Securities, HDFC Securities
 (d) ICICI Direct, HDFC Securities, Upstox
 (e) None of the above
151. The GST collection of which months in 2023 stood at ₹1.72 lakh crore, the second highest-ever?
 (a) July (b) August
 (c) September (d) October
 (e) June
152. The Department of Consumer Affairs, under the Ministry of Consumer Affairs, Food and Public Distribution, has added how many new districts to the existing network of 288 districts where the mandatory BIS hallmarking laws for gold are in force?
 (a) 45 (b) 56
 (c) 76 (d) 80
 (e) 96
153. Which Public Sector Enterprise has been conferred with the prestigious 'CSR Times Awards 2023' in 'Swachh Bharat' category for the 'Revival and operation of Municipal Solid Waste Management Plant, at Karsada, Varanasi'?"
 (a) NHPC (b) ONGC
 (c) Indian Oil (d) Bharat Petroleum
 (e) NTPC
154. Which Pension Fund Manager has launched a first-of-its-kind NPS Preference Index that highlights the preference of NPS amongst different consumer groups?
 (a) HDFC Pension Management
 (b) ICICI Prudential Pension Fund Management
 (c) UTI Retirement Solutions Ltd
 (d) SBI Pension Fund
 (e) LIC Pension Fund
155. India has won a gold medal in men's squash team event after beating which team by 2-1 at the Asian Games 2023?
 (a) Nepal (b) Malaysia
 (c) Pakistan (d) South Korea
 (e) Thailand

Answer Key & Solutions

1. (a) **Statement I:** The research indicating the benefits of coffee was funded by major coffee companies.
 This statement suggests a potential conflict of interest and questions the credibility of the research. If the study

was funded by entities that stand to benefit from positive results, the findings may be biased. Therefore, this statement weakens the given passage by casting doubt on the validity of the recommendation.

Statement II: High consumption of coffee can lead to increased anxiety and sleep disorders.

While this points out negative side effects of drinking coffee, it does not directly refute the claim that coffee prevents type

2 diabetes. It raises concerns but doesn't weaken the specific conclusion of the passage.

Statement III: Not everyone can tolerate coffee due to its caffeine content.

This highlights that the recommendation may not be suitable for everyone but doesn't challenge the effectiveness of coffee in preventing diabetes. It doesn't weaken the passage's main claim.

Statement IV: Studies show that diet and exercise are more effective in preventing type 2 diabetes.

This introduces alternative methods for prevention but doesn't contradict the claim that coffee can prevent diabetes. It doesn't weaken the passage; it merely offers other options.

Statement V: Excessive caffeine intake can have adverse health effects.

Similar to Statement II, this mentions negative side effects but doesn't directly dispute the effectiveness of coffee in preventing type 2 diabetes. It doesn't weaken the passage's conclusion.

Hence, Only Statement I directly weakens the passage by questioning the credibility of the research.

2. (b) **Statement I:** Recent clinical trials have revealed that NovaGen's vaccines are more effective against emerging viral strains compared to those produced by Zenith.

If NovaGen's vaccines are more effective, this is a significant advantage.

Effectiveness against emerging strains is crucial for a national immunization program.

This would persuade the Ministry to consider NovaGen over Zenith.

Statement II: Zenith Pharmaceuticals has been accused of unethical practices, including data manipulation in clinical trials, by international health organizations.

Accusations of unethical practices and data manipulation are serious concerns.

This undermines trust in Zenith's vaccines and their effectiveness.

The Ministry would be reluctant to renew the contract with a company under such allegations.

Statement III: NovaGen Biotech's production facilities have failed multiple regulatory inspections due to non-compliance with safety standards.

Non-compliance with safety standards is a major red flag.

This raises concerns about the safety and reliability of NovaGen's vaccines.

The Ministry would be hesitant to award the contract to NovaGen under these circumstances.

Statement IV: The cost of vaccines offered by Zenith is significantly lower than those offered by NovaGen, easing

the financial burden on Country A's healthcare budget.

Lower costs are a strong incentive to continue with Zenith.

Financial constraints might make the Ministry favor Zenith despite other concerns.

This would support renewing the contract with Zenith.

Statement V: Country A has committed to international agreements that prioritize the procurement of the most advanced and effective vaccines available.

The commitment to procure the most advanced and effective vaccines aligns with choosing NovaGen if their vaccines are more effective (as per Statement I).

This obligation would push the Ministry towards selecting the supplier with superior vaccine technology.

Therefore, the combination of Statements I, II, and V most strongly persuades the Ministry to not renew with Zenith and award the contract to NovaGen.

3. (a) **Statement I:** Conduct an independent audit of the company's products to verify their environmental claims and publicly release the findings.

An independent audit adds credibility to the company's environmental claims.

Publicly releasing the findings demonstrates transparency.

This can help restore consumer trust if the products meet environmental standards.

Statement II: Reduce the prices of all products to match competitors, even if it means compromising on quality.

Lowering prices may attract price-sensitive customers.

However, compromising on quality can damage the brand's reputation further.

This could lead to a loss of loyal customers who value quality and environmental impact.

Statement III: Invest in research and development to innovate new, genuinely eco-friendly products that meet higher environmental standards.

Innovating new products can differentiate EcoLife from competitors.

Meeting higher environmental standards strengthens the brand's core value proposition.

This can attract environmentally conscious consumers and restore market position.

Statement IV: Launch an aggressive marketing campaign highlighting the company's commitment to the environment and discrediting competitors' products.

Highlighting commitment to the environment is positive.

However, discrediting competitors can be seen as negative campaigning.

Without addressing the underlying issues, marketing may not be effective.

Could lead to legal challenges if claims about competitors are unfounded.

Statement V: Collaborate with environmental organizations to develop certification programs that validate the eco-friendliness of products in the market. Collaboration with reputable organizations enhances credibility.

Certification programs can set industry standards and benefit consumers.

This can help rebuild trust and position EcoLife as a leader in eco-friendly products.

Solutions (4-8):

	I	II	III	IV
I	B	A	T	S
II	L	I	N	E
III	C	O	R	D
IV	M	E	L	T

4. (d) B

5. (b) $L(12) + I(9) + N(14) + E(5)$
 $= 12 + 9 + 14 + 5 = 40$

6. (c) CORD

7. (d) Row III Sum: $C(3) + O(15) + R(18) + D(4) = 40$

Column II Sum: $A(1) + I(9) + O(15) + E(5) = 30$

Difference: $40 - 30 = 10$

8. (a) T

9. (b) **Statement I:** Several environmental organizations have warned that the introduction of genetically modified seeds may lead to a loss of biodiversity and could harm native plant species.

Impact: Raises environmental concerns about GM seeds.

Relevance: Potential ecological risks could lead to opposition from environmental groups and the public.

Severity: While significant, these concerns can be mitigated with regulations and controlled implementation.

Conclusion: This statement weakens the plan but may not be the most serious threat.

Statement II: Studies have shown that advanced irrigation systems proposed under AgriBoost are not suitable for the varied topography and soil conditions prevalent in many regions of Country X.

Impact: Suggests that a key component of the policy is impractical in many areas.

Relevance: If irrigation systems are unsuitable, they won't improve productivity and could waste resources.

Severity: This seriously undermines the effectiveness of the policy in achieving its goals.

Conclusion: This statement seriously weakens the plan.

Statement III: A significant portion of the farming community lacks awareness and training on the use of GM seeds and advanced irrigation technologies, leading to potential misuse and crop failures.

Impact: Points out a lack of necessary skills and knowledge among farmers.

Relevance: Without proper training, farmers may not benefit from the policy, and it could result in adverse outcomes.

Severity: This poses a serious risk to the success of the policy.

Conclusion: This statement seriously weakens the plan.

Statement IV: Countries that have implemented similar agricultural modernization policies have witnessed a substantial increase in crop yields and farmers' incomes over the past decade.

Impact: Provides evidence that similar policies have been successful elsewhere.

Relevance: Supports the government's plan by showing potential positive outcomes.

Severity: Strengthens the plan.

Conclusion: This statement strengthens the government's plan.

Statement V: The low-interest loans provided under AgriBoost have stringent eligibility criteria, which most small-scale farmers in Country X cannot meet.

Impact: Indicates that many farmers will be excluded from financial support.

Relevance: Limits the policy's reach and effectiveness, especially among those who may need it most.

Severity: Seriously undermines the inclusivity and equity of the policy.

Conclusion: This statement seriously weakens the plan.

10. (a) **Statement I:** A significant portion of the population in Country Y lives below the poverty line and is more concerned with meeting basic needs like food and shelter than participating in fitness programs.

This statement suggests that the target population is unlikely to participate in the program because their primary focus is on survival needs.

If the majority of people are struggling with basic necessities, they may not prioritize fitness, rendering the program ineffective.

This directly challenges the assumption that making fitness facilities accessible will lead to widespread adoption of healthier lifestyles.

Conclusion: This statement most weakens the Ministry's decision.

Statement II: Private gyms and fitness centers are offering advanced facilities and personalized training, attracting those

who can afford them and leaving public centers underutilized.

This statement indicates that wealthier individuals prefer private facilities.

However, the government's program aims to provide access to those who cannot afford private gyms.

The existence of private gyms does not necessarily weaken the decision to provide public facilities for the broader population.

Conclusion: This statement does not significantly weaken the Ministry's decision.

Statement III: There is a cultural stigma in certain regions of Country Y against women participating in public fitness activities, which may limit the program's effectiveness.

This statement highlights a potential challenge in certain regions.

While it may limit participation among women in those areas, it does not negate the potential benefits of the program nationwide.

The program can still be effective overall, and strategies can be implemented to address cultural sensitivities.

Conclusion: This statement weakens the decision to some extent but not as significantly as Statement I.

Statement IV: Similar programs in neighboring countries have successfully reduced obesity rates and improved public health over the past five years.

This statement provides evidence that such programs can be successful.

It supports the Ministry's decision rather than weakens it.

Conclusion: This statement strengthens the Ministry's decision.

Statement V: The government's budget allocation for the FitnessFirst program is limited, leading to concerns about the quality and maintenance of the public fitness centers.

Budget constraints could affect the quality of facilities.

However, a limited budget does not necessarily mean the program will fail; it may still have a positive impact.

The statement suggests challenges but does not directly undermine the decision to launch the program.

Conclusion: This statement somewhat weakens the decision but not as significantly as Statement I.

Solution (11-15):

Step I: Numbers: The square of the first digit is done. One is added to the second digit and third digit. One is subtracted from the fourth digit. (Operations are performed from left to right.)

Word: Vowels are placed in the alphabetical order and then consonants are placed according to the alphabetical order. (operations are performed according to the first alphabet of given words)

Step II: Numbers: First even numbers are placed and then odd numbers are placed in ascending order (operations are performed from left to right.). Zero is eliminated in this step.

Word: The succeeding alphabet is considered in Vowels and Preceding alphabets are considered in consonants.

Step III: Numbers: Add all the even digits separately and Add all the odd digits separately.

Word: Eliminate vowels and repeated alphabets.

Step IV: Number: Add all numbers.

Word: Place the alphabet in alphabetical order from left to right within the word.

Step V: If the number of alphabets is odd then add the number to the number preceding it otherwise subtract it.

Input: 8536 GRACE 2847 MOVIE 6183 BLEND 7312 SMILE 4765 POWER

Step I: 64645 AECGR 4956 EIOMV 36292 EBDNL 49421 EILMS 16874 EOPRW

Step II: 44665 BFBFQ 4659 FJPLU 22639 FACMK 24419 FJKLR 46817 FPOQV

Step III: 205 Q 1014 FJPL 1012 FCMK 1010 FJKLR 188 FPQV

Step IV: 7 Q 6 FJLP 4 CFKM 2 FJKLR 17 FPQV

Step V: 8 2 0 7 13

11. (c) 22639

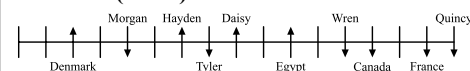
12. (d) FCMK

13. (d) 30

14. (a) FJKLR

15. (b) 13

Solutions (16-19):



16. (d) The same number of persons are sitting to the left of Wren as to the right of the one who is sitting second to the left of Hayden.

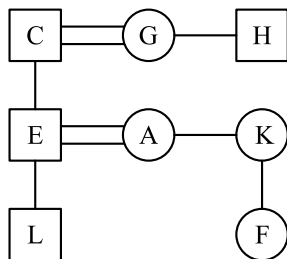
17. (e) The one who is sitting second to the left of Daisy is sitting fourth to the right of the person who is sitting second to the right of Morgan.

18. (c) The one who is from Egypt and Wren are sitting second to the right of each other.

19. (b) Seven persons are sitting between Daisy and Quincy.

Solutions (20-23):

Years	Age	Student
1953	71	G
1957	67	H
1963	61	C
1982	42	E
1985	39	K
1988	36	A
2015	09	F
2018	06	L



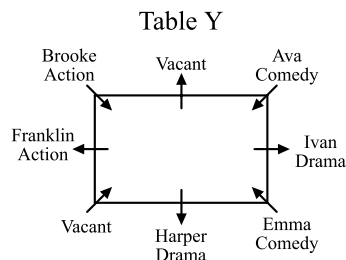
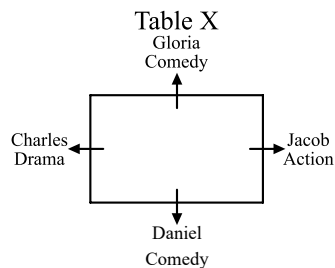
20. (e) A is born immediately after K and is 36 years old. Hence, the one who is born immediately after K age is a perfect Square.
21. (c) C is born in 1963.
22. (a) K is the mother of F.
23. (b) The sum of the age of E and K is 81 years.

Solutions (24-27):

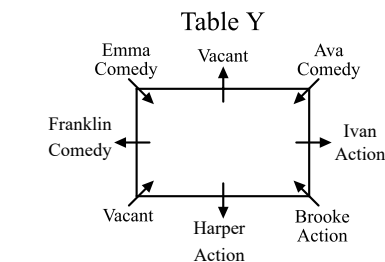
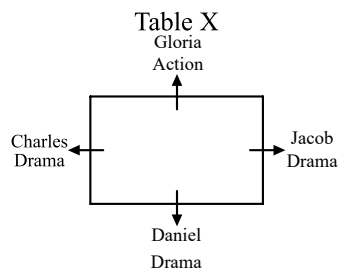
P1	P2	P3	P4
C (6 AM), D (7AM)	F, G, E	B, K, A	J, H

24. (b) Platform P2 and P3 have more than two buses.
25. (e) Bus B's destination is P3. Bus F's destination is ahead of Bus D's destination.
26. (a) Bus A, Bus B, Bus K have a destination as P3.
27. (c) Bus D reached the platform P1 at 7 AM.

Solutions (28-31):



After the rearrangement:



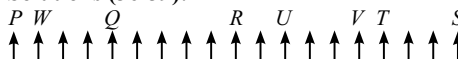
28. (a) Jacob is sitting on Table X. Hence, Only II (Emma, Fraklin, and Jacob are sitting on the table Y.) is false statement.
29. (c) The person who is sitting third to the right of person who is sitting immediately right of Fraklin likes Action novels.
30. (b) Gloria and Daniel are the immediate neighbors of Jacob.
31. (d) All the persons except Ivan are the neighbors of Vacant seat. Hence, Ivan does not belong to the group.

Solutions (32-35):

Year	Month	Person	Watches
1964 age 56	June	D	Tissot
	July	E	Noise
1977 age 43	June	F	Omega
	July	A	Titan
1982 age 38	June	G	Timex
	July	B	Casio
1991 age 29	June	H	Amazfit
	July	C	Citizen

32. (e) Both (a) and (b)
33. (d) July 1964
34. (d) C
35. (a) 33 years

Solutions (36-39):



S.No.	Cartons	Colour
8	R	Violet
7	T	Yellow
6	V	Crimson
5	P	Orange
4	U	Maroon
3	Q	Indigo
2	S	Black
1	W	Blue

36. (d) Nineteen
37. (c) Violet
38. (c) Eight
39. (c) Three
40. (a) From-I

Days	Case-1		Case-2	
	Person	Floors	Person	Floors
Monday	G			
Tuesday	E/		G	
Wednesday	I/	8 TH	E	8 TH
Thursday	H		I	
Friday	E/		H	
Saturday	I/			

From-II

Days	Case-1		Case-2	
	Person	Floors	Person	Floors
Monday		1st		
Tuesday	I			1st
Wednesday			I	
Thursday				
Friday		5th		
Saturday	J		J	5th

Now from I, II

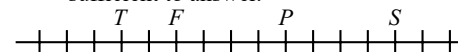
Days	Person	Floors
Monday	G	
Tuesday	E	1 st
Wednesday	I	8 st
Thursday	H	
Friday	F	
Saturday	J	5 st

So, clearly E has office on 1st floor attend office on Tuesday

41. (d) T sits fourth from the left end. Only one person sits between F and T.

From statement I,

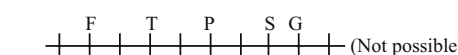
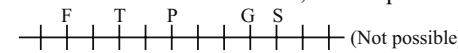
Three persons sit between F and P. At least two persons sit between P and T, so P must sit at 10th from left. As S sits 4th to the right of P, so S sits on 14th from left. And less than 3 persons sit to the right of S. At least 5 persons sit between P and the person sitting at the extreme end, so 16 persons sit in the row. Statement I is sufficient to answer.



From statement II,

P sits 4th to the right of F, so P either sits on 6th or 10th from left. Now only two persons sit between P and G, who is not the neighbour of F, so G sits on either 9th or 13th from left.

Case-I: S either sits on 8th or 10th from left and only two persons sit to the right to S. So, total numbers will be either 10 or 12, which is less than 13. So, it is not possible.



Case-II: S either sits on 12th or 14th from left and only two persons sit to the right of S, so the number of persons in the row is either 14 or 16. Also, one person sit between T and F and T is 4th from left end.

— T — F — P — S — G — (Possible)

— T — F — P — G — S — (Possible)

So, statement II is not sufficient to answer the question.

Solutions (42-45):

Floors	Flat 1	Flat 2	Flat 3
5	Q ₹30,000 (61)	X ₹80,000 (82)	V ₹1,00,000 (96)
4	—	W ₹85,000 (72)	P ₹40,000 (48)
3	Z ₹10,000 (63)	—	R ₹90,000 (76)
2	T ₹50,000 (90)	U ₹70,000 (60)	—
1	M ₹20,000 (69)	S ₹60,000 (85)	Y ₹95,000 (55)

42. (c) 159 kg

43. (d) The one who earns Rs. 70,000

44. (b) Flat number 3 of floor number 3.

45. (a) Two floors between X and T.

46. (d) The correct rearrangement of the words is BAC/BCA.

Obfuscated: Rendered unclear or unintelligible, often deliberately.

Bewildered: Confused, perplexed, or deeply puzzled.

47. (d) The correct rearrangement of the words is ABC/ACB.

Factions: Small, organized groups within a larger organization or body, often with conflicting interests.

Overtly: Openly, without concealment or secrecy.

Covertly: Secretly or in a way that is not openly acknowledged.

48. (e) The correct rearrangement is ABC/BAC/CAB

Methodical: Performed or arranged in a systematic way; characterized by careful and orderly planning.

Deliberate: Done consciously and intentionally; carefully considered.

Ponderous: Having great weight

49. (b) The correct rearrangement is ABC/BAC.

50. (b) The correct rearrangement of the words is ABC/ACB.

Esoteric: Intended for or understood by only a small number of people with special knowledge; obscure or mysterious.

Ubiquitous: Present, appearing, or found everywhere; widespread.

51. (b) This option maintains the flow of the conversation and aligns with the content in paragraphs I and III, where the interviewee is asking about the team and projects.

52. (d) The chosen option (d) complements paragraph (I), where it is discussed that the country's economic growth has been driven by factors like domestic demand, strategic investments in key sectors, and innovation. Option (d) reinforces the idea that economic success thrives on diversification, emphasizing the importance of moving beyond traditional economic sectors and investing in technology and services to enhance economic resilience and create a dynamic job market. It aligns well with the theme of economic growth and diversification presented in paragraph (I).

53. (a) This Option introduces the topic of access to quality education in the context of reducing inequality and promoting sustainable development, aligning well with the themes presented in the previous paragraphs.

54. (c) This option introduces the theme of water scarcity and its impact on agriculture.

55. (d) This sentence reinforces the idea that baking is not just about satisfying taste buds but also about bringing happiness and nourishing the soul, aligning with the passage's theme.

56. (a) The error in the sentence is related to subject-verb agreement. The subject "Scientists" is a plural noun, and the verb "has" is in the singular form, which is incorrect. The verb should agree in number with the plural subject. Thus, the verb "has" should be replaced with "have".

57. (e) The given sentence is grammatically correct.

58. (a) The error in the sentence is in part (A). To maintain consistency, the correct form should be "One of the students" in part (A) so that it aligns with the rest of the sentence.

59. (b) The error in Part (B) use "between" instead of "among." "Between" is used to describe a relationship between two distinct entities—in this case, air pollution levels and type 2 diabetes incidence. "Among" implies a connection within a group, which does not fit the context of this sentence.

60. (e) The given sentence is grammatically correct.

61. (d) The error lies in Part (D) due to a lack of parallelism. Both activities in the list should have the same structure. To ensure parallelism, "browse websites" should be changed to "browsing websites."

62. (a) **Meticulous:** Showing great attention to detail; extremely careful and precise in work or tasks.

Scrutinize: To examine or inspect closely and thoroughly, often with critical attention to details.

Complex: Having many interconnected parts; intricate or complicated in structure.

The other options are not relevant to the sentence's context

63. (c) **Pragmatic:** Dealing with problems or situations in a practical and realistic way, rather than being guided by theory or dogma.

Practical: Concerned with the actual implementation or application of ideas or solutions in a real-world context.

Idealistic: Characterized by pursuing high or noble ideals, often without regard to practical considerations, and often having a vision of perfection.

The other options are not relevant to the sentence's context

64. (d) **Eloquent:** Fluent or persuasive in speaking or writing; able to express thoughts and ideas clearly and effectively.

Captivated: Attracting and holding the interest and attention of someone, often because of being fascinating or emotionally engaging.

Conveyed: Communicated or expressed a message or information to others, often through words, actions, or gestures.

The other options are not relevant to the sentence's context

65. (b) **Tenacious:** Holding fast or firm, characterized by persistence and determination in pursuing a goal or objective.

Breakthrough: A significant advancement, discovery, or achievement, often implying a notable and positive development.

Ultimately: Refers to something happening or being resolved in the end or eventually. The other options are not relevant to the sentence's context

66. (c) **Conundrum:** A confusing or difficult problem or puzzle, often one that requires careful thought to solve.

Limited: Restricted in quantity, quality, or extent, indicating that there is not an abundant supply or available resources are constrained.

Consideration: The process of thinking carefully about something, or the act of taking various factors or aspects into account when making a decision or forming a judgment.

The other options are not relevant to the sentence's context

67. (e) **Quixotic:** Extremely idealistic, unrealistic, and impractical, often to the point of being foolish.

Wavered: To show hesitation or uncertainty in one's commitment or determination.

The other options do not fit the sentence's context

68. (b) Refer to the lines, "In an epoch characterized by the ceaseless generation and utilization of data across sundry devices, systems, and applications, the peril of clandestine data access is exponentially exacerbated."

Option (b) emphasizes that the continuous generation and utilization of data across various devices and applications contribute to the potential for unauthorized access.

69. (c) Refer to the lines, "The repercussions of data breaches transcend mere pecuniary ramifications, precipitating deleterious effects on an organization's reputational standing and instigating punitive measures from regulatory bodies."

The correct answer is (c) Both (I) and (II).

70. (c) Refer to the lines, "Cyber threat actors, whether individuals or groups, employ various methods, such as malware or phishing emails, to breach corporate networks."

Option (c) cannot be considered as a data breach in the realm of cyber threats because standard network monitoring is a legitimate and essential cybersecurity practice. In contrast, options (a), (b), and (d) describe actions or events that could potentially result in a data breach

71. (e) The passage does not explicitly provide information on the specific strategies or methods that businesses employ to navigate compliance with stringent data protection regulations.

72. (b) The most suitable word to fill in the blank is "absconds" Here are the meanings of the provided words:

(a) **Elucidates:** To make something clear or explain it in a way that is easy to understand; to clarify or illuminate.

(b) **Absconds:** To leave hurriedly and secretly, typically to avoid detection or arrest; to depart suddenly and covertly.

(c) **Vilifies:** To speak or write about someone in an abusive or defamatory manner; to slander or criticize harshly.

(d) **Exacerbates:** To make a situation, problem, or negative feeling more intense, severe, or worse; to worsen or aggravate.

(e) **Ruptured:** To break or burst open, especially in a forceful or violent way; to tear apart or rupture.

73. (b) The tone of the passage is "Informative".

The passage provides information and analysis about data breaches, discussing their nature, methods employed by cyber threat actors, and the broader implications for organizations. It lacks the sarcastic

or didactic tone and doesn't convey an overtly pessimistic or analytical stance.

74. (b) The most suitable word with a similar meaning to "Ominous" is: (b) threatening Here are the meanings of the provided words:

(a) **Propitious:** Giving or indicating a good chance of success; favourable.

(b) **Threatening:** Expressing an intention to cause harm or danger; indicating a potential danger or harm.

(c) **Cacophony:** A harsh, discordant mixture of sounds; a jarring, unpleasant noise.

(d) **Ubiquitous:** Present, appearing, or found everywhere; being or seeming to be everywhere at the same time.

(e) **Disparate:** Essentially different in kind; distinct in quality or character; markedly distinct or different.

75. (a) Refer to the lines, "While voice recognition and speech recognition are often used interchangeably, they have distinct meanings. Voice recognition identifies the speaker, whereas speech recognition assesses the content of what is said."

The accurate representation of the specific focus of voice recognition in comparison to speech recognition is best encapsulated by option (a): "Voice recognition predominantly centers on discerning the speaker's identity rather than appraising the uttered content."

76. (c) Refer to the lines, "This digital database of words or syllables is then compared to signals during pattern recognition."

The correct answer is (c) Pattern recognition, as described in the passage, involves comparing the digital database of words or syllables to incoming signals during the processing phase. This comparison is crucial for the system to recognize and interpret spoken words.

77. (d) Refer to the lines, "Voice recognition involves analyzing speech through models like the hidden Markov model or recurrent neural networks. The former breaks down spoken words into phonemes, while the latter uses previous outputs to influence current inputs, improving capabilities and accuracy with increased data."

The hidden Markov model breaks down spoken words into smaller units called phonetic components or phonemes. This detailed analysis helps the voice recognition system better understand and recognize the unique patterns in the way people speak, making the overall voice recognition capabilities more accurate and effective.

78. (e) The correct answer is (e) Both (b) and (d).

Refer to the lines, "As more users engage with voice recognition, the wealth of data

generated enhances the capabilities and accuracy of these systems, indicating a promising trajectory for the technology's future." The passage implies that the wealth of data is crucial for the positive development of voice recognition technology because it provides a diverse range of speech patterns, accents, and variations (option b). This diverse dataset enriches the system's database, contributing to its adaptability and accuracy. Furthermore, the passage also suggests that the abundance of data enhances the capabilities of voice recognition systems to accurately interpret and respond to a wide array of voices and linguistic nuances (option d). Therefore, both (b) and (d) accurately capture the reasons presented in the passage.

79. (a) The most suitable word to fill the blank is: (a) advent Meaning of words:

(a) **Advent:** The arrival or creation of something notable or significant, often used to refer to the beginning of a new era or development.

(b) **Oblivious:** Lacking awareness, attention, or memory of something; not conscious or mindful.

(c) **Conservation:** The act of preserving, protecting, or efficiently using resources, often in the context of the environment or natural resources.

(d) **Adversity:** Difficulties, misfortune, or unfavourable circumstances; facing challenges or hardship.

(e) **Mundane:** Ordinary, commonplace, or lacking excitement; pertaining to the ordinary aspects of everyday life.

80. (d) The most appropriate choice that can substitute the three highlighted words in the passage in the same order is: (d) mandate, elevated, indispensable.

Mandate: an official order or commission to do something elevated: raised to a higher position; increased in rank or status indispensable: absolutely necessary; essential.

Solutions (81-84):

According to the question,

$$6p + 8p = 420$$

$$\text{Or, } 14p = 420$$

$$\text{Or, } p = 30$$

$$\text{Also, } 10q - 8q = 48$$

$$\text{Or, } 2q = 48$$

$$\text{Or, } q = 24$$

For Store 'A':

$$\text{Revenue generated from debit card} = 6p$$

$$= 6 \times 30 = 180 \text{ millions}$$

$$\text{Revenue generated from cash transactions} = 12.5q = 12.5 \times 24 = 300 \text{ millions}$$

$$\begin{aligned} \text{Total revenue generated worldwide} \\ = 300 + 180 = 480 \text{ millions} \end{aligned}$$

Stores	Revenue generated from debit card (in millions)	Revenue generated from cash transactions (in millions)	Total revenue generated worldwide (in millions)	Rating
A	180	300	180 + 300 = 480	2
B	270	180	270 + 180 = 450	2
C	330	240	330 + 240 = 570	1
D	150	192	150 + 192 = 342	3
E	240	288	240 + 288 = 528	1

81. (c) Required ratio = $(180 + 270) : 450$
 $= 450 : 450 = 1 : 1$

82. (d) Required revenue = $(0.4 \times 150) + (0.75 \times 192) = 60 + 144 = 204$ million

83. (e) Both store 'A' and 'B' are given the rating '2'.

84. (a) Total revenue generated worldwide for the stores which got rating '1'
 $= (570 + 528) = 1098$ million
 Required percentage
 $= \frac{1098 - 300}{300} \times 100 = 266\%$

85. (e) For city 'Delhi':
 Percentage of cars in Delhi
 $= (100 - 25)\% = 75\%$
 So, total number of vehicles in Delhi = $600 \div 0.75 = 800$
 So, number of motorcycle in Delhi
 $= 800 - 600 = 200$
 Number of cars in Jaipur = $0.3b$
 So, number of motorcycle in Jaipur
 $= b - 0.3b = 0.7b$
 So, $a = 0.7b - 0.3b = 0.4b$
 So, $5a = 2b$

86. (c) According to question,
 $0.75 \times \text{total number of vehicles in city 'Pune'} = 900$
 $\Rightarrow \text{Total number of vehicles in city 'Pune'} = 900 \div 0.75 = 1200$
 So, number of motorcycle in city 'Pune' = $1200 - 480 = 720$
 So, required percentage
 $= \left(\frac{720}{1200} \right) \times 100 = 60\%$

87. (d) Let total number of vehicles in city 'Jaipur' be $10p$
 So, number of cars in city 'Jaipur'
 $= 10p \times 0.3 = 3p$
 And number of motorcycle in city 'Jaipur'
 $= 10p - 3p = 7p$

And number of cars in city 'Punjab'

$$= 3p \times \left(\frac{9}{5} \right) = 5.4p$$

And number of motorcycle in city 'Punjab' = $(7p + 340)$ or $(7p - 340)$

When number of motorcycle in city 'Punjab' is $7p + 340$, then $7p + 340 + 5.4p = 900$

$$\text{Or, } 12.4p + 340 = 900$$

$$\text{Or, } 12.4p = 560$$

So, $p \sim 45.16$ (Since, 'p' has to be an integer, we have to discard this value of 'p')

$$\text{So, } 7p - 340 + 5.4p = 900$$

$$\text{Or, } 12.4p = 1240$$

$$\text{Or, } p = 100$$

So, total number of vehicles in city 'Jaipur' = $100 \times 10 = 1000$

Solutions (88-90):

$$z^2 + 2z - 224 = 0 \Rightarrow z^2 + 16z - 14z - 224 = 0$$

$$\Rightarrow z(z + 16) - 14(z + 16) = 0$$

$$\Rightarrow (z + 16)(z - 14) = 0 \Rightarrow z = -16, 14$$

(i) The values given in the given pie chart in the form of degree or percentage form cannot be negative.

$$\text{So, } z = 14$$

$$(19z - 4y) = 42$$

Put the value of 'z' in the above equation.

$$(19 \times 14 - 4y) = 42 \Rightarrow (266 - 4y) = 42, 4y = 224$$

$$y = 56$$

(1) The total number of laptops sold by all the five stores (A, B, C, D and E) taken together is 7200.

Number of laptops sold by store A

$$= 26\% \text{ of } 7200 = \frac{26}{100} \times 7200 = 1872$$

Number of laptops sold by store B

$$= (3y - 16) = (3 \times 56 - 16) = (168 - 16) = 152$$

$$360^\circ = 7200 \Rightarrow 1^\circ = 20$$

Number of laptops sold by store C

$$= 56^\circ = 56 \times 20 = 1120$$

Number of laptops sold by store D

$$= (2z - 5)\% \text{ of } 7200$$

$$= (2 \times 14 - 5)\% \text{ of } 7200 = 23\% \text{ of } 7200$$

$$= \frac{23}{100} \times 7200 = 1656$$

Number of laptops sold by store E

$$= (8.5z + 1)^\circ = (8.5 \times 14 + 1)^\circ$$

$$= (119 + 1)^\circ = 120^\circ = 120 \times 20 = 2400$$

Store	Laptops sold
A	1872
B	152
C	1120
D	1656
E	2400

88. (e) Number of non gaming laptops sold by store E = $(100 - 38)\% \text{ of } 2400$

$$= 62\% \text{ of } 2400$$

$$= \frac{62}{100} \times 2400 = 1488$$

89. (d) The average number of laptops sold by store A, C, E and B together

$$= \frac{(1872 + 1120 + 2400 + 152)}{4}$$

$$= \frac{5544}{4} = 1386$$

The average number of laptops sold by store B and D together

$$= \frac{(152 + 1656)}{2} = \frac{1808}{2} = 904$$

$$\text{Required answer} = 1386 - 904 = 482$$

90. (b) $(100 - 75)\%$ of the number of laptops sold by store F = $2400 - 1656 = 744$

$$25\% \text{ or } \frac{1}{4} \text{ of the number of laptops sold}$$

$$\text{by store F} = 744$$

Number of laptops sold by store F

$$= 744 \times 4 = 2976$$

Number of laptops unsold by store F

$$= \frac{2976}{62} \times 75 = 3600$$

91. (e) Number of laptops sold by store A on

$$\text{MRP} = \left(\frac{5}{6} \right) \text{ of } 1872 = 1560$$

Number of laptops sold by store A on discount = $1872 - 1560 = 312$

$$\text{Now, } 68000 \times 1560 + (100 - a)\% \text{ of } 68000 \times 312 = 122204160$$

$$\Rightarrow 68000 \times 1560 + 680 \times (100 - a) \times 312 = 122204160$$

$$\Rightarrow 106080000 + 212160(100 - a)$$

$$= 122204160$$

$$\Rightarrow 212160(100 - a)$$

$$= 122204160 - 106080000$$

$$\Rightarrow 212160(100 - a) = 16124160$$

$$\Rightarrow (100 - a) = 76 \Rightarrow a = 100 - 76 = 24$$

$$92. (b) 50\% = \frac{1}{2} \text{ and } 16.66\% = \frac{1}{6}$$

Time taken by Mohani and Aman together to complete the work = $15 \times 2 = 30$ days

Time taken by Ritu and Kiran together to complete the work = $6 \times 6 = 36$ days

Let the total work be 360 units.

Let the efficiency of, Ram = r, Mohani

$$= m, \text{ Aman} = a, \text{ Sam} = s, \text{ Kiran} = k$$

$$(r + m) \times 24 = (m + a) \times 30 = (a + s) \times 24$$

$$= (r + k) \times 36 = (s + k) \times 20 = 360$$

$$r + m = 15, m + a = 12, a + s = 15, r + k$$

$$= 10, s + k = 18$$

$$(m + a) + (a + s) - (r + m) - (s + k) + (r +$$

$$k) = 12 + 15 - 15 - 18 + 10$$

$$\Rightarrow 2a = 4 \Rightarrow a = 2$$

Hence required time taken by Bittu to complete 40% of work

$$= \frac{(360 \times 0.4)}{(1.5 \times 2)} = 48 \text{ days}$$

93. (a) Simple interest = (Sum $\times$ rate of interest $\times$ time period in years) $\div$ 100

$$\text{So, simple interest earned} = \{(a + 1000) \times 20 \times 3\} \div 100 = ₹0.6a + 600$$

And compound interest earned

$$= (a - 4000) \times \left\{1 + \left(\frac{20}{100}\right)\right\}^3 - (a - 4000)$$

$$= (a - 4000) \times (1.728 - 1)$$

$$= (a - 4000) \times 0.728 = ₹0.728a - 2912$$

Here, two cases are possible;

Case (I): simple interest $>$ compound interest

$$\text{So, } 0.6a + 600 = 0.728a - 2912 + 1080$$

$$\text{Or, } 2432 = 0.128a$$

$$\text{So, } a = 19000$$

Case (II): simple interest $<$ compound interest

$$\text{So, } 0.6a + 600 + 1080$$

$$= 0.728a - 2912$$

$$\text{Or, } 4592 = 0.128a$$

$$\text{So, } a = 35875$$

For statement I:

Since, 'a' can take two values therefore, statement I is true.

For statement II:

Again, 'a' = 19,000 or 35875 which are greater than 10,000

So, statement II is false.

For statement III:

Since, 'a' can be even or odd therefore statement III is false.

94. (e) Amount received by 'X'

$$3000t + \left\{ \frac{6000t}{[6000t + (t-3)8000] + (t+3)5000} \right\} \times (81000 - 3000t)$$

$$= 3000t + \frac{\{6t \times (81000 - 3000t)\}}{(19t - 9)}$$

$$= 3000t + \frac{486000t - 18000t^2}{19t - 9}$$

Amount received by 'Z'

$$= \frac{5000(t+3)}{\{[6000t + (t-3)8000 + (t+3)5000] \times (81000 - 3000t)\}}$$

$$= \frac{\{5(t+3) \times (81000 - 3000t)\}}{(19t - 9)}$$

$$\text{Now, } 3000t + \frac{486000t - 18000t^2}{(19t - 9)}$$

$$= 25000 + \frac{\{(5t+15)(81000 - 3000t)\}}{(19t - 9)}$$

$$\Rightarrow 57000t^2 - 27000t + 486000t - 18000t^2 = 475000t - 225000 + 405000t - 15000t^2 + 1215000 - 45000t$$

$$\Rightarrow 54000t^2 = 376000t + 990000$$

$$\Rightarrow 27t^2 - 188t - 495 = 0$$

$$\Rightarrow 27t^2 + 55t - 243t - 495 = 0$$

$$\Rightarrow 27t(t-9) + 55(t-9) = 0$$

$$\Rightarrow (27t + 55)(t - 9) = 0 \Rightarrow t = 9 \text{ or } -\frac{55}{27}$$

$$\text{So, } t = 9$$

Profit share of 'Y'

$$= \frac{8 \times (t-3)}{19t-9} \times (81000 - 3000t)$$

$$= \left\{ \left(\frac{8 \times 6}{162} \right) \right\} \times 54000 = ₹16000$$

95. (a) According to question,

$$(a + 5) = 0.6(2b - 5)$$

$$\text{Or, } 1.2b - a = 8 \quad \dots (i)$$

$$\text{Also, } 1.25(2.5a - 5) = 2b - 5$$

$$\text{Or, } 1.6b - 2.5a = -1 \quad \dots (ii)$$

On solving equation (i) and (ii), we get:

$$a = 10 \text{ and } b = 15$$

$$\text{Number of white bands} = (a + b)$$

$$= 10 + 15 = 25$$

$$\text{Number of orange bands} = (3b - a)$$

$$= (3 \times 15 - 10) = 35$$

$$\text{Number of yellow bands} = (b - a) = 5$$

$$\text{Total number of rubber bands in box}$$

$$= a + b + 3b - a + b - a = 5b - a$$

$$= 5 \times 15 - 10 = 65$$

$$\text{Required probability} = \frac{25_{C_2}}{65_{C_2}}$$

$$= \frac{(25 \times 24)}{(65 \times 64)} = \frac{15}{104}$$

96. (a) According to question,

Quantity I:

Let the time taken by tap 'P' and 'Q' to fill the water tank together be 'a' hours.

According to question,

$$\frac{1}{a+4} + \frac{1}{15} = \frac{1}{a}$$

$$\Rightarrow a \times (15 + a + 4) = 15 \times (a + 4)$$

$$\Rightarrow 19a + a^2 = 15a + 60 \Rightarrow a^2 + 4a - 60 = 0$$

$$\Rightarrow a^2 + 10a - 6a - 60 = 0$$

$$\Rightarrow a(a+10) - 6(a+10) = 0$$

$$\Rightarrow (a+10)(a-6) = 0$$

$$\text{So, } a = 6 \text{ or } a = -10$$

Since, time cannot be negative, therefore $a = 6$.

And, time taken by tap 'P' to fill the alone = $6 + 4 = 10$ hours.

$$\text{So, } p = 10$$

$$\text{So, } 25p = 10 \times 25 = 250$$

$$\text{So, Quantity I} = 250$$

Quantity II:

Required number of ways

$$= (7!) \div (3! \times 2! \times 2!) = 210$$

So, Quantity II = 210

So, Quantity I $>$ Quantity II.

97. (e) **Quantity I:**

$$7y^2 - 476y + 8092 = 0$$

$$\Rightarrow 7(y^2 - 68y + 1156) = 0$$

$$\Rightarrow y^2 - 68y + 1156 = 0$$

$$\Rightarrow y^2 - 34y - 34y + 1156 = 0$$

$$\Rightarrow y(y - 34) - 34(y - 34) = 0$$

$$\Rightarrow (y - 34)(y - 34) = 0$$

$$\Rightarrow y = 34$$

According to question,

$$\frac{1}{3} (100 - y)\% \text{ of } (z + 3300) = 12826$$

$$\Rightarrow \frac{1}{3} (100 - 34)\% \text{ of } (z + 3300) = 12826$$

$$\Rightarrow (z + 3300) \times 22\% = 12826$$

$$\Rightarrow (z + 3300) \times 22 = 12826 \times 100$$

$$\Rightarrow (z + 3300) = 58300 \Rightarrow z = 55000$$

$$\text{So } 2.5\% \text{ of 'z'} = \frac{(2.5 \times 55000)}{100} = 1375$$

Quantity II: $1.2P^2 = 2268750$

$$\Rightarrow P^2 = 1890625$$

$$\text{Value of 'P'} = 1375, -1375$$

So, Quantity I $\geq$ Quantity II.

Solutions (98-99):

$$\text{Series I: } 60 + (2^4 - 2^3) = 60 + (16 - 8)$$

$$= 60 + 8 = 68$$

$$68 - (3^4 - 3^3) = 68 - (81 - 27) = 68 - 54 = 14$$

$$14 + (4^4 - 4^3) = 14 + (256 - 64)$$

$$= 14 + 192 = 206$$

$$206 - (5^4 - 5^3) = 206 - (625 - 125)$$

$$= 206 - 500 = -294$$

$$-294 + (6^4 - 6^3) = -294 + (1296 - 216)$$

$$= -294 + 1080 = 786$$

$$786 - (7^4 - 7^3) = 786 - (2401 - 343)$$

$$= 786 - 2058 = -1272$$

$$\text{So } P = 206$$

$$\text{Series II: } (P - 12) = 206 - 12 = 194$$

$$194 + (2^4 - 2^3) = 194 + (16 - 8)$$

$$= 194 + 8 = 202 = Q$$

$$202 - (3^4 - 3^3) = 202 - (81 - 27)$$

$$= 202 - 54 = 148 = R$$

$$148 + (4^4 - 4^3) = 148 + (256 - 64)$$

$$= 148 + 192 = 340 = S$$

$$340 - (5^4 - 5^3) = 340 - (625 - 125)$$

$$= 340 - 500 = -160 = T$$

$$160 + (6^4 - 6^3) = 160 + (1296 - 216)$$

$$= 160 + 1080 = 1240 = U$$

$$1240 - (7^4 - 7^3) = 1240 - (2401 - 343)$$

$$= 1240 - 2058 = -818 = W$$

98. (c) $R = 148 = \frac{340 + 920}{2}$

99. (e) First term of third series = $\frac{S + U}{2}$
 $= \frac{1260}{2} = 630$

$630 + (2^4 - 2^3) = 630 + (16 - 8)$
 $= 630 + 8 = 638$

$638 - (3^4 - 3^3) = 638 - (81 - 27)$
 $= 638 - 54 = 584$

$584 + (4^4 - 4^3) = 584 + (256 - 64)$
 $= 584 + 192 = 776$

$776 - (5^4 - 5^3) = 776 - (625 - 125)$
 $= 776 - 500 = 276$

The sum of values of 'T' and 'W'
 $= -160 + (-1138) = -1298$

This is not the multiple of 9. So the statements (i) is not true.

The fourth term of the third series = 776

This is not an odd number. So the statement (ii) is not true.

The difference between the value of 'Q' and the second term of the third series
 $= 638 - 202 = 436$

So the statement (iii) is not true.

100. (d) Distance travelled by Vasu = 780 km
 Distance travelled by Sanjeet = 540 km
 Required difference = 780 - 540
 $= 240$ km

101. (c) There are following two ways to travel from P to R:

(i) P Q S R

(ii) P S R

Distance will be lesser in case of P - S - R.

Required time = $\frac{780}{60} + \frac{900}{45}$
 $= 13 + 20 = 33$ hours

102. (b) Let the train fare from P to Q be ₹x km
 So, the taxi fare Q to S = 1.25x/km

According to the question,
 $x \times 360 + 1.25x \times 600 = 19980$
 $\Rightarrow 360x + 750x = 19980$
 $\Rightarrow 1110x = 19980 \Rightarrow x = 18$

103. (e) Time required to travel from P to S

$= \frac{780}{60} = 13$ hours

There are the following two ways to travel from R to S:

(i) R P S

Time required = $\frac{420}{40} + \frac{780}{60}$
 $= 10.5 + 13 = 23.5$ hours

(ii) R Q S

Time required = $\frac{720}{40} + \frac{600}{50}$
 $= 18 + 12 = 30$ hours

Minimum possible sum of the time
 $= 13 + 23.5 = 36.5$ hours

104. (b) Since the digit at tens place of PQR is half of X therefore X is an even number.

Possible values of X are 2, 4, 6 and 8.

Now, the product of PQR and X, i.e., 2634, is divisible by only 2 and 6. But dividing 2634 by 2 will give a four-digit number.

So, the only possible value of X = 6

$\therefore \text{PQR} = \frac{2634}{6} = 439$

(i) Unit digit of 439 is 9 (True)

(ii) 439 is a prime number. (True)

(iii) $4 + 3 + 9 = 16 \neq 18$ (False)

Solution (105-106):

Here, diagonal of cube = diameter of sphere

$\sqrt{3}C = 2R \Rightarrow 3C^2 = 4R^2$

$\Rightarrow \frac{R^2}{C^2} = \frac{3}{4} \Rightarrow R = \frac{\sqrt{3}C}{2}$

105. (d) Required ratio = $\frac{4 \times \frac{22}{7} \times R^2}{4 \times C^2}$

$= \frac{22}{7} \times \frac{3}{4} = \frac{33}{14}$

106. (e) Radius of cone = $\sqrt{3} \times \frac{\sqrt{3}}{2} C = \frac{3}{2} C$

According to question,

$\frac{1}{3} \times \frac{22}{7} \times \left(\frac{3}{2} C\right)^2 \times H = 3168$

$\Rightarrow C^2 = \frac{1344}{H}$

Solution (107-108):

Factors of X are 13, 5 and (A - 8).

Since 1 will be one of the factors of X therefore,

$A - 8 = 1 \Rightarrow A = 9$

So, $X = 13 \times 5 \times 1 = 65$

$B = 31 + 65 = 96$

$\sqrt[n]{9}$ will be integer for $n = 2$

Now, the series is:

3, 31, 96, C, 439, 783, 1296

$3 + 3^3 + 1 = 31$

$31 + 4^3 + 1 = 96$

$96 + 5^3 + 1 = 222 = C$

$222 + 6^3 + 1 = 439$

$439 + 7^3 + 1 = 783$

$783 + 8^3 + 1 = 1296$

107. (a) $A + C - n = 9 + 222 - 2 = 229$

108. (c) $B + 34 = 96 + 34 = 130$

LCM of (130, 65) = 130

Solutions (109-111):

Putting $x = 5$ in equation A:

$(5 - 2)^2 = (-3 \times 5^2) + 2^2 + 25 \times 5 - P$

$\Rightarrow P = 45$

Now, A : $(x - 2)^2 = (-3x^2) + 2^2 + 25x - 45$

$\Rightarrow 4x^2 - 29x + 45 = 0$

Sum of roots = $-\left(-\frac{29}{4}\right) = 7.25$

$\therefore$ Other root of the equation = $7.25 - 5$

$= 2.25 = \frac{9}{4}$

$B : \left(10y^2 - 3^2y + \frac{2}{3}\right) \times 3 + 10y = 0$

$\Rightarrow 30y^2 - 17y + 2 = 0 \Rightarrow y = \frac{2}{5}, \frac{1}{6}$

109. (d) Required answer

$= \frac{7 \times 45}{45 - 24} \times 0.2 \times 45 - 91$
 $= 135 - 91 = 44 = P - 1$

110. (c) Required answer = $\frac{9}{4} \times 11 = 24 \frac{3}{4}$

111. (a) Smallest root of equation A = $\frac{9}{4}$

Largest root of equation B = $\frac{2}{5}$

112. (a) **Quantity I:**

$\frac{(2a^3 + 2b^3)(a - b)}{(2a^2 - 2ab + 2b^2)(a^2 - b^2)} = \frac{(a^3 + b^3)}{(a^3 + b^3)} = 1$

Quantity II: Since the value of Q is negative and less than -1, so -2Q is positive and greater than 1.

Also, minimum value of P can be 0, therefore $(P - 2Q) > 1$

$\therefore$ Quantity I < Quantity II

113. (b) Possible values for N are 104, 401, 203, 302 and 500.

The product of any prime number with even number will give an even number. Since M is an odd number, values for N can be odd, i.e., 401 and 203.

M is an odd number, and its unit digit cannot be 5. So its unit digit can be 1, 3 and 7.

Four-digit prime multiples of 203 which are less than 3400 are 1421 and 2233.

$\therefore$ Four-digit prime multiples of 401 which are less than 3400 are 1203 and 2807.

Since the sum of digits of M = 8 therefore M = 1421 and N = 203

I. Product of the digits of M = 8

Sum of the digit of N = 5

$8 - 5 = 3$ (True)

II. Difference between the hundred place digits of M and N = $4 - 2$

$= 2$ (True)

III. $M + N = 1421 + 203 = 1624$

$\neq 1729$ (false)

Solutions (114-115):

Let the efficiency of A be 3 units/day.

$\therefore$ Efficiency of C = 2 units/day

50% of efficiency of B = 2 units/day

So, efficiency of B = $2 \times 2 = 4$ units/day

$\frac{4}{3}$ of efficiency of D = 4 units/day

So, efficiency of D = 3 units/day

Total work = $14 \times (3 + 4 + 2 + 3) = 168$ units

$$\text{So, } x = \frac{168}{3+4} = 24$$

$$y = \frac{168}{3+3} = 28$$

114. (e) Time required by P and Q together to complete work 'Y' = $24 - 9$

= 15 days

Time required by P and R together to complete work 'Y' = $28 - 4 = 24$ days

Time required by P, Q and R together to

$$\text{complete work 'Y'} = \frac{3 \times 24}{0.5 \times 28 - 8} = \frac{72}{6}$$

= 12 days

Let the total work be 120 units (LCM of 15, 24 and 12).

$$\text{Efficiency of (P + Q)} = \frac{120}{15} = 8 \text{ units/day}$$

$$\text{Efficiency of (P + R)} = \frac{120}{24} = 5 \text{ units/day}$$

$$\text{Efficiency of (P + Q + R)} = \frac{120}{12}$$

= 10 units/day

Efficiency of Q = $10 - 5 = 5$ units/day

Efficiency of R = $10 - 8 = 2$ units/day

$$\text{Required percentage} = \frac{5-2}{5} \times 100$$

= 60%

115. (c) $j \times (x + 4) = k \times (y - 8)$

$$\Rightarrow j \times (24 + 4) = k \times (28 - 8)$$

$$\Rightarrow j = \frac{5}{7}k \quad \dots(i)$$

$$\text{Also, } k \times (y - 8) = (y + 2) \times (0.5k + 3.5)$$

$$\Rightarrow k \times (28 - 8) = (28 + 2) \times (0.5k + 3.5)$$

$$\Rightarrow 20k = 15k + 105 \Rightarrow k = 21$$

$$\text{So, } j = \frac{5}{7} \times 21 = 15 \text{ [using eq. (i)]}$$

$$\text{Required time} = \frac{21 \times (28 - 8)}{21 - 15}$$

= 70 days

116. (c) The chairman of the 8-member high-level panel formed by the government to address the concept of "One nation, One election" is Ram Nath Kovind.

The committee was constituted to examine the feasibility of holding simultaneous elections to the Lok Sabha and state assemblies.

The panel includes other prominent members like Home Minister Amit Shah, Congress leader Adhir Ranjan Chowdhury, and former Finance Commission Chairman NK Singh.

117. (a) The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) has joined the One CGIAR integrated partnership.

The One CGIAR partnership includes the CGIAR System Organisation and 12 One CGIAR research centres.

The partnership aims to create a unified approach to addressing the challenges posed by the climate crisis in food, land, and water systems.

118. (d) The Union Cabinet, chaired by the Hon'ble Prime Minister, has approved a Scheme for Viability Gap Funding (VGF) for the development of Battery Energy Storage Systems (BESS).

The scheme aims to develop 4,000 MWh of BESS projects by 2030-31, with financial support of up to 40% of the capital cost as budgetary support in the form of VGF.

The initial outlay for the scheme is Rs.9,400 crore, with a budgetary support of Rs.3,760 crore, which signifies the government's commitment to sustainable energy solutions.

119. (a) Small Industries Development Bank of India (SIDBI) inks MoU with India Post Payments Bank (IPPB) for development and financing of informal micro enterprises in rural and remote areas of the country by leveraging strengths of respective institutions. IPPB will assist SIDBI's customers in adopting digitization through merchant onboarding systems and providing UPI and QR-based solutions.

120. (b) The Psyche mission by NASA aims to study a metal-rich asteroid of the same name, located in the main asteroid belt between Mars and Jupiter. This asteroid, named 16 Psyche, is believed to be composed mostly of metallic iron and nickel, similar to the Earth's core. By studying Psyche up close, scientists hope to gain insights into the early formation of planetary bodies and the processes that led to the differentiation of metallic cores within rocky planets like Earth.

121. (d) The Reserve Bank of India introduced the Digital Payments Index in January 2021 with the base year as March 2018.

The DPI assesses the level of digitalization in payments nationwide and showcases the growth of different digital payment methods.

122. (a) Indian-origin scientist Joyeeta Gupta has been awarded the Spinoza Prize, the highest Dutch science award, for her work in the field of climate change.

She is recognized for her outstanding, pioneering, and inspiring scientific work focusing on a just and sustainable world.

The award includes 1.5 million euros to be used for scientific research and knowledge utilization activities.

Gupta is a professor of Environment and Development at the University of Amsterdam and the co-chair of the Earth Commission.

123. (a) India's first 3D-printed post office building was inaugurated in Bengaluru by Union minister Ashwini Vaishnaw.

The post office building covers an area of 1,000 square feet and was constructed using 3D printing technology with a robotic printer.

Larsen & Toubro Limited was responsible for the construction, with technical guidance provided by the Indian Institute of Technology (IIT), Madras.

124. (b) PhonePe has launched a new shopping app called "Pincode" on the ONDC platform focused on hyperlocal commerce for customers in Bangalore, with expansion to other cities planned

Pincode aims to digitally connect consumers with neighborhood stores for online ordering, discounts, and instant refunds and returns, promoting local shopkeepers and sellers

The app will feature both national brands and locally manufactured products, offering a wider selection than competitors, providing stimulus for local retailers and the entire local commerce ecosystem, including MSME's and farmers

125. (a) RBI approved SBI Mutual Fund to acquire 9.99% of the paid-up share capital in IndusInd Bank.

The acquisition allows SBI MF to hold 9.99% voting rights in the bank.

SBI MF has been advised by RBI to acquire the major shareholding in the bank by October 10, 2024.

SBI MF must ensure that its aggregate holding in the bank does not exceed 9.99% of the paid-up share capital or voting rights.

126. (c) Axis Bank has launched the first numberless credit card in India in partnership with fintech startup Fibe (previously known as EarlySalary).

The Fibe Axis Bank Credit Card details can be accessed through the Fibe app.

The numberless credit card provides added security as it does not display the card number, expiry date, or CVV on the card plastic, reducing the risk of identity theft and unauthorized access to card details.

127. (d) State Bank of India (SBI) has introduced a new Customer Service Points (CSP) functionality for hassle-free enrolment in social security schemes.

Customers only need their Aadhaar cards to enrol in schemes like Pradhan Mantri

Jeevan Jyoti Bima Yojana, Pradhan Mantri Suraksha Bima Yojana, and Atal Pension Yojana.

The new feature aims to remove barriers to financial security and empower every section of society.

This advancement is expected to widen the coverage of social security schemes.

128. (c) While 25% of the I-CRR maintained would be released on September 9, 2023, another 25% would be released on September 23, 2023 and the balance 50% would be released on October 7, 2023.

129. (b) The International Solar Alliance (ISA) is a global intergovernmental organization aimed at promoting the widespread adoption of solar power.

It was initiated by India and is currently led by the country.

The ISA seeks to provide a framework for collaboration between countries to facilitate the deployment of solar energy technologies and to promote research and development in the field.

The alliance is expected to play a significant role in accelerating the transition to a sustainable energy future by increasing the use of clean, renewable solar power.

130. (b) The International Monetary Fund (IMF) has revised India's economic growth forecast for FY24 to 6.3% from 6.1% earlier.

Retail inflation in India is expected to rise to 5.5% in FY24 before easing to 4.6% in FY25.

The Reserve Bank of India (RBI) has projected consumer price index (CPI)-based inflation for the current fiscal year at 5.4% and GDP growth at 6.5%.

The IMF also expects India's current account deficit to remain at 1.8% of GDP in FY24 and FY25.

131. (e) The Sacred Ensembles of the Hoysala, temples in Chennakeshava Temple at Belur, Hoysaleswara Temple at Halebidu, and Keshava Temple at Somanathapura, have been added to the UNESCO World Heritage list.

This marks the 42nd UNESCO World Heritage Site in India and comes after Rabindranath Tagore's Santiniketan received the same recognition.

The Hoysala temples were India's nomination for consideration as World Heritage for the year 2022-2023.

132. (a) Sikkim recently experienced a Glacial Lake Outburst Flood (GLOF). The South Lhonak Lake, a glacial lake located at an altitude of 17,000 feet in the state's northwest, experienced a rupture as a result of continuous rainfall.

Consequently, water was discharged into the downstream regions, causing flooding in the Teesta River and impacting four districts of Sikkim: Mangan, Gangtok, Pakyong, and Namchi, as reported by the Sikkim State Disaster Management Authority (SSDMA).

133. (b) The name of the cutting edge rocket defense system in Israel is the "Iron Dome".

It is a missile defense system developed by Israel and Rafael Advanced Defense Systems, primarily intended to protect the country against short-range rockets launched from Gaza and the West Bank.

134. (b) India, the world's fourth largest refiner, plans to add 56.6 million tonnes per annum (mtpa) of crude oil refining capacity in the next seven years.

84% of the capacity expansion will be through brownfield expansion, while 16% will be through greenfield expansion.

India currently has a cumulative refining capacity of almost 254 mtpa, or a little over five million barrels per day (mb/d).

The Ministry of Petroleum and Natural Gas (MoPNG) states that India will add nine mtpa of refining capacity through greenfield expansion.

135. (d) Longpi pottery originates from the state of Manipur in India.

It is a traditional craft practiced by the Tangkhul Naga tribe in the Ukhrul district of Manipur.

Longpi pottery is known for its unique craftsmanship and use of natural materials such as serpentine rock and weathered rock.

TRIFED's showcase at the G-20 highlights India's rich tribal heritage and craftsmanship.

136. (c) NASA astronaut Frank Rubio has made history by returning to Earth after an extraordinary 371 days in space, shattering previous records.

Originally scheduled for a modest six-month mission aboard the International Space Station, Rubio's journey turned into an epic adventure that earned him a place in the annals of space exploration.

However, the ultimate record for the longest duration spent in space remains with the late Russian cosmonaut Valeri Polyakov, who completed an astounding 437 continuous days aboard Russia's Mir space station between January 1994 and March 1995.

137. (a) The high-speed ferry will cover a distance of about 60 nm (110 Km) between Nagapattinam and KKS in around 3.5 hours, depending on sea conditions.

To commence the service, the Centre supported the Tamil Nadu Maritime Board in upgrading facilities at the Nagapattinam port.

138. (c) The Finance Commission recommends the distribution of the net proceeds of taxes between the Union government and the state governments.

This ensures a fair and equitable sharing of tax revenues, taking into account the fiscal capacities and needs of the states.

The Finance Commission determines the principles and quantum of grants-in-aid to states that require financial assistance.

139. (a) Athletics Federation of India (AFI) partners with HSBC India to support women athletes in India.

The collaboration focuses on identifying promising girls from the National Inter District Championships (U-14 and U-16 categories).

This initiative aims to enhance women's participation in track and field disciplines and elevate their performance levels on the global stage.

140. (a) The Nobel Peace Prize is uniquely awarded in Oslo, Norway, unlike the other Nobel Prizes which are presented in Stockholm, Sweden.

This tradition dates back to the will of Alfred Nobel, the founder of the Nobel Prizes, who specified in 1895 that the Peace Prize be awarded by a Norwegian committee.

The award is presented by the Norwegian Nobel Committee, appointed by the Norwegian Parliament (Storting).

141. (e) In a tokenisation transaction, all key parties in the payment ecosystem are involved to ensure secure and seamless card-based digital payments.

The customer initiates the transaction using a tokenised version of their card instead of actual card details.

The merchant accepts the payment and sends it through the payment gateway.

The merchant's acquirer processes the transaction and routes it to the issuing bank.

The issuer (customer's bank) validates and authorises the tokenised transaction.

142. (b) The sport of billiards was not included in the Olympic program for the 2028 Games.

The Olympic Committee announced in 2020 that billiards would be dropped from the Paris Games due to a lack of global support.

The decision brought an end to the sport's Olympic career, which began in 1902 in London. Despite this, billiards remains a popular recreational activity worldwide.

- 143. (a)** Primary market is a market wherein corporates issue new securities for raising funds generally for long term capital requirement. The companies that issue their shares are called issuers and the process of issuing shares to public is known as public issue.
- 144. (c)** The Central Board of Direct Taxes (CBDT) has instructed its officials not to carry out scrutiny of startups recognised by the Department for Promotion of Industry and Internal Trade (DPIIT) for angel tax provisions as amended in Budget 2023-24.
The CBDT has directed its officers to refrain from scrutinizing the angel tax provisions for start-ups that have received recognition from the DPIIT.
This directive comes in response to concerns raised by many start-ups regarding scrutiny notices for angel tax.
- 145. (c)** The Valley Queen Heritage Train is Rajasthan's only heritage train service. It replicates the design of a 150-year-old steam engine, providing passengers with a nostalgic experience.
Rajasthan is preparing to introduce its exclusive heritage train service, the Valley Queen Heritage Train, which will commence its journey from Marwar Junction to Khamlighthat starting on October 5. Prime Minister Narendra Modi virtually inaugurated this unique rail service from Jodhpur.
- 146. (c)** The Government of India (GOI), through Nationalisation took over the shares of 55 Indian insurance companies and the undertakings of 52 insurers carrying on general insurance business. General Insurance Corporation of India (GIC) was formed in pursuance of Section 9(1) of GIBNA.
- 147. (e)** Enhanced Access and Service Excellence (EASE) evolved over four years from FY19 through FY22, bringing improvements to Public Sector Banks. EASENext program introduced EASE 5.0 'Common reforms agenda' for Public Sector Banks by Minister of Finance and Corporate Affairs Nirmala Sitharaman in New Delhi via video conference.
- The EASE 5.0 initiatives cover diverse themes including business growth, profitability, risk, customer service, operations, and capability building.
- 148. (e)** The Direct Benefit Transfer (DBT) scheme aims to transfer subsidies directly to the beneficiaries through their linked bank accounts.
It has been implemented in various government schemes to ensure efficient and transparent delivery of subsidies and benefits. Some of the schemes where DBT is implemented include:
(a) PM Awas Yojana
(b) National Food Security Mission
(c) Pradhan Mantri Krishi Sinchai Yojana
(d) PM KISAN
- 149. (b)** Google introduced its AI-powered search feature to India and Japan after its initial availability in the United States.
The revamped search experience was unveiled at the annual Google I/O event in May.
The AI-powered search feature aims to enhance the quality and relevance of search results.
- 150. (a)** Zerodha is the top stock broker in India with the most active clients since 2019. Zerodha is followed by Upstox, Angel One, ICICI Direct and Groww.
There are 20 stock brokers in India with over 1 lakh active customers.
The year-wise list of top stock brokers includes the latest number of active clients with the broker.
- 151. (d)** GST collections in October 2023 stood at ₹1.72 lakh crore, the second highest-ever.
The collections are 13 per cent higher than the ₹1.52 lakh crore collected in October 2022.
The highest-ever revenue from Goods and Services Tax (GST) was recorded in April 2023 at ₹1.87 lakh crore.
The average gross monthly GST collection in the FY 2023-24 now stands at ₹1.66 lakh crore, 11 per cent higher than the year-ago period.
- 152. (b)** The government has recently expanded the mandatory hallmarking of gold in India.
The Department of Consumer Affairs, under the Ministry of Consumer Affairs, Food and Public Distribution, has added 56 new districts to the existing network of 288 districts where the mandatory hallmarking laws for gold are in force.
- 153. (e)** NTPC won the CSR Times Awards 2023 in the 'Swachh Bharat' category for reviving and operating the Municipal Solid Waste Management Plant in Karsada, Varanasi.
The award was received by Shri Neeraj Kapoor (GM - CSR/R&R/LA) and Shri Promod Kumar Sahu (AGM - CSR) on 21st August 2023 in Delhi.
- 154. (a)** HDFC Pension, one of India's leading Pension Fund Managers, has been at the forefront in creating consumer awareness amongst Indians about retirement planning using the National Pension System (NPS).
October 1st is marked as NPS Diwas. To coincide with the day, HDFC Pension has launched a first-of-its-kind NPS Preference Index that highlights the preference of NPS amongst different consumer groups.
The NPS is a government-backed retirement savings program designed to provide financial security to Indian citizens during their retirement years.
- 155. (c)** India won a gold medal in men's squash team event at the Asian Games by defeating Pakistan 2-1.
Abhay Singh led the Indian team to victory with a thrilling win in the final round over Noor Zaman.
Saurav Ghosal and Mahesh Mangaonkar were part of the victorious team, repeating their success from the Incheon 2014 Games.
The Indian team secured a hard-fought 2-0 victory over 2018 champions Malaysia in the semi-finals.

